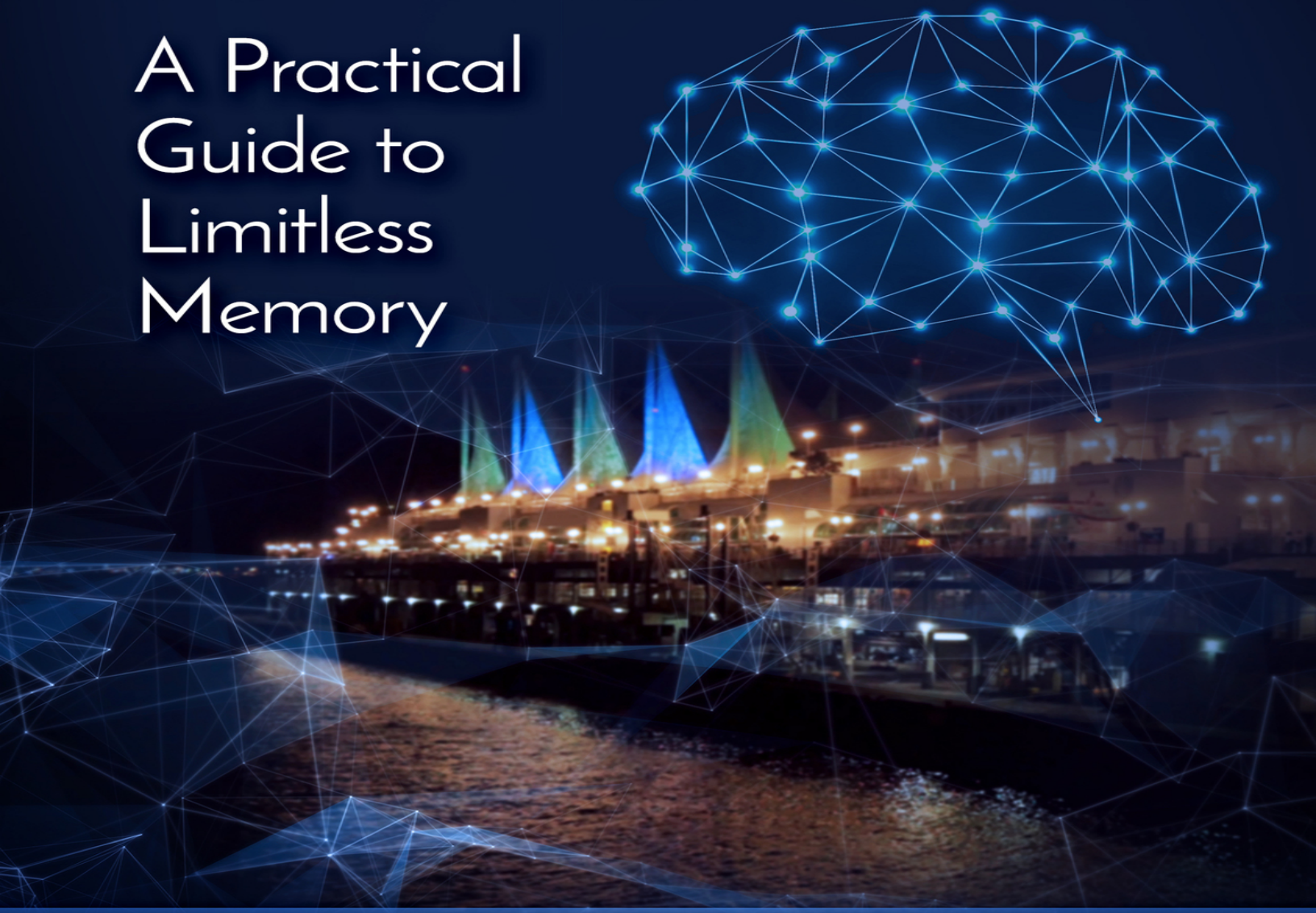
PHOTOGRAPHIC MEMORY FOR BEGINNERS

A Practical Guide to Limitless Memory



DANE KRAUSS

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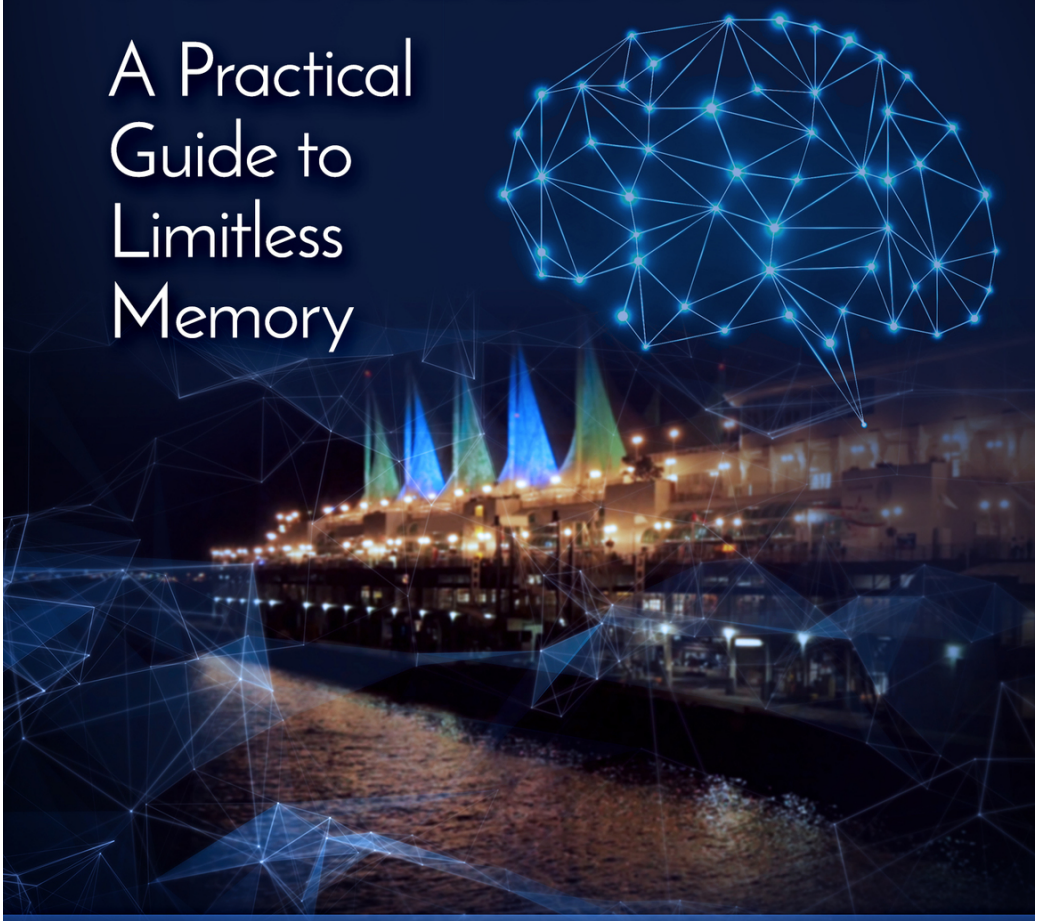

DANE KRAUSS

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Photographic Memory for Beginners

A Practical Guide to Limitless Memory

Book 2 of the Mind Improvement
For Beginners Series

Dane Krauss

Foreword

Have you been feeling frustrated because you keep forgetting things? Are you not getting your expected productivity when you work because you can't focus and there too many distractions? You are not alone. Absentmindedness, mental blocks and forgetfulness are common memory failures that usually cause impairments of performance and productivity. These are memory issues easily addressed by training your memory.

This book will show you exactly how to train your memory and keep your brain performances in best shape. Read on and discover how this book gives us all valuable insights to keeping your memory young and fit. The author, gives you the following no-brainer pointers to take care of your memory, improve your brain's performance, introduce yourself to mnemonic methods and principles, and more of a variety of valuable notions about memory improvement.

Introduction

These days, you probably find there are more and more things you have to keep in mind. As a result, you will need to have a good memory. Not everyone has a good memory - most people will tend to forget one thing or another from time to time. If this is the case for you, you should not worry as there are several ways to improve memory. You can find many solutions which will be help you to improve your memory.

The best way to go about this is to combine all the available. By doing this, you will find you will have a higher chance of achieving a greater level of memory improvement. The higher the level of improvement you get, the higher the chances are of your using more of your memory.

Improvement will result in several advantages. One of these advantages will be the fact you will be able to memorize names of the people who you have just met. This can be a very great experience, especially for people who deal with a lot of clients. When you have improved your memory to this level, you will have a greater chance of providing your clientele with exceptional service and as a result, they may become repeat customers.

This will lead to increased production in your company. The second important benefit of memory improvement will be that you will be able to keep important numbers in your memory. For every day use, it would be important to have your identification number, bank account number; credit card numbers, phone numbers for your next of kin, etc. in your head. This is because it will be very convenient in case anything unexpected happens.

An example of something unexpected happening would be an accident. In this case, you may be able to give your rescuers the phone numbers for your next of kin so they can be informed of your plight. If you happen to be a public speaker, you can benefit a lot from memory improvement. This improvement can help you be able to memorize speeches in a short time irrespective of their length - so if you are called on unexpectedly, you can rise to the occasion!

Most public speakers read from or refer to speeches placed in front of them. This is because they are in need of memory improvement. A public speaker who reads from a speech that has been handed to him / her needs memory

improvement tactics. Those who speak from their minds have a good memory.

Apart from these benefits, there are others associated with memory improvement. You are able to remember vividly events that have been in the news or other media. This means if such a topic were to arise in a conversation, you are able to quote articles from newspapers and contribute to the conversation as a result of your memory improvement. These reasons are why it is advisable for the average person to look for memory improvement methods.

Memories are crucial to us as human beings because, in so many ways, it helps shape who we are. They work as an internal biography of our lives, documenting the things that happened to us, what we've done, and make the stories we tell about ourselves. Memory serves as the sum total of what we remember and helps us learn from previous experiences. Consider for a moment how many times in a day you rely on your memory to help you with passwords, finding directions, or greeting someone on their birthday. Human memory makes all these things possible and more. The retrieval of relevant information is something we all do on a regular, daily basis, and whether we realize it or not, we rely so much on memories more than we ever accomplish. Some memories are brief, while others can be stored for a long time, and possibly throughout our whole lifetime. In order to make new memories, our brain processes information and stores them for later use. And while this is usually done far from our awareness, this formed memory is kept ready for retrieval when the time comes for us to need them.

Owing to the fact that memories are so important to us, its loss can be very frustrating and can have devastating effects on our daily lives. And while age-related memory loss is inevitable, there are many different ways to preserve the integrity of this process through memory enhancement exercises. This book will explore the many methods and techniques that can help you improve your memory, enhance brain function, and will help you develop proven ways to preserve your memory.

Chapter 1: A Brief History of Memory Systems

The history of the memory systems dates back to ancient times. Back then, there were note-taking devices and everything was noted by memory. It was the memory techniques and systems that helped the early Romans and Greek speakers and orators to deliver their speeches accurately without the help of notes.

In the middle of the 20th century, the idea that memory was composed of distinct systems became a point of experimental research. In 1980, evidence from experimental animals, normal people, and amnesic patients established there was a palpable distinction between the kind of memory that can be accessed to conscious recollection, and another kind that cannot be accessed through the same process.

The following inquiries shifted the findings to a biological view that memory consisted of multiple separate systems. The brain tissues, like the hippocampus and related structures such as the cerebellum, the amygdala, and the neostriatum, are supposed to support the separate systems of memory. Various inquiries traced the development of this idea, and provided more up-to-date standpoint on how these brain structures work to support behavior.

The Formation of Memories

Newly born cells help “stamp” memories according to the time they occurred. The new cells do not record exact dates. Instead, they encode memories that occurred around the same time; we have the inclination of whether the other memories happened before or after this event. Neuroscientists studied that when the same neurons are active during two specific events, a link or connection between the two memories is formed.

These links or connections are called synapses. When brain cells encode the sensory information from our life events into our long-term memory, the brain combines the information and categorizes them by linking them to similar data.

The memories are more likely to last if we visit them time and again, thus retracing the synapses and strengthening the connections.

Memory encoding starts being possible once a baby’s pre-frontal cortex is developed. The prefrontal cortex is the area of the brain located near the forehead. It becomes completely developed by the 24th month. By then, it’s capable of forming synapses just like adults.

Chapter 2: Memory and the Brain

Thanks to our memory, we remembered seeing that line before and all we had to do was go to our quote file and there it was. Either both men did utter the same sentence (highly unlikely) or it's a mere typographical error (more likely).

A bad memory is the last thing we'd want. We scold ourselves for forgetting to buy an important ingredient after we've come back from the supermarket. We scold ourselves because we forgot our wedding anniversary and hence got a severe reprimand and a cold shoulder in return.

We scold ourselves because we forgot our dental appointment and then got fined $50.00 for a no-show. Unfortunately, even dentists act like policemen nowadays. One fines you for speeding, and the other penalizes you for not showing up. If we had to pay fines for everything we forgot, we'd be bankrupt by now. That does not bode well for our financial future; nor does it bode well for the future of our mind.

Brief Visit to Memory Lane

What exactly is this thing called memory?

It depends from what perspective we look at it. In Psychology, memory is a human being's ability to store and remember bits and pieces of information and eventually to use such information when the need arises. Memory used to fall under the field of philosophy, but in the early 20 century, it was integrated into cognitive psychology. Today, memory now properly belongs to a branch of science that combines cognitive psychology and neuroscience. Thus, it is now under the realm of cognitive neuroscience .

Given that the study of memory in cognitive neuroscience is complex, we shall not attempt to dissect it and examine its numerous facets. This complexity has resulted in a multi-classification system wherein memory has been classified by duration, by information type and by temporal direction.

For our purposes, we will take the first classification – by duration – and explain it briefly before we deal with ways to improve our memory. Under memory by duration, we have three types: Sensory memory, short-term memory and long-term memory.

According to Wikipedia, sensory memory corresponds to the first instant that a person or object is noticed. This observation sometimes moves into the sensory store, and qualifies as short-term memory. Sensory memory means that the observation or perception lasts from milliseconds to seconds. Short-term memory, in turn, corresponds to memory that occurs from seconds to minutes.

Immediate interaction with things, objects and persons constitute our sensory memory as they are information produced by our five senses. Touching, smelling or seeing are faculties that enable us to remember more easily.

Long-term memory is memory that is stored and then retrieved after days

and years.

An example may help us understand this system of classification better: supposing someone gives us his number. We remember it for a few seconds – maybe even up to a minute or two, but soon forget it. This is short-term memory at work. On the other hand, we remember certain phone numbers we use frequently. Our baby sitter's number, the pharmacist's, our husband's office – these numbers are stored in our long-term memory.

Memory Improvement Techniques

This section will cover general techniques that generate benefits for us in the short and long-term, we describe some exercises you can do. You don't have to adopt all of them. Choose a couple or at least one exercise you can do consistently to develop your memory. One exercise is better than none. As "If you start today and practice, practice, practice, soon your memory will be as a retentive as flypaper (though what gets stuck to it will, with luck, be more useful)."

It's a pity our memories don't act the same way as computers. A computer-like brain would be a boon to our daily lives, wouldn't it? The human brain, although not as dazzling a performer as a Pentium IV, is far more complex; however, while it may not be able to spew out chunks of data in minutes, the human mind has been responsible for how humanity and civilization evolved. This makes memory a very valuable asset and tool. It is not mechanical; it's the stuff we're made of.

Before describing specific memory improvement techniques, we'll take a quick refresher on two aspects of memory: learning and concentration.

Each person has his own way of learning and concentrating. These two stages are prerequisites to memory building. Learning is the acquisition of data and actual skills, while concentration is the mind's ability to focus well on something with the least amount of distraction.

Learning

Individuals learn in three ways: looking, listening and doing. There are individuals who rely mostly on sight, others on their sense of hearing and still others who learn by doing. Certain measurements exist to gauge one's most predominant learning style. We'll deal briefly with some of these practical tests:

For instance, after watching a movie, which part do you remember most the dialogue, the action sequences, or the things you did, like driving to the cinema, buying the movie pass and popcorn? If you answered "dialogue," that makes you a listener. If you answered "action sequences," you are a looker, and if you answered the "things you did," that makes you a doer.

Another example: If you moved to a new city, how would you find your way around: (a) ask people for directions, (b) buy a map, or (c) walked around the neighborhood to familiarize yourself with the layout of the city? If you answered (a), you are a listener, if you picked (b), you're a looker and if you chose (c), you are a doer .

Of course we need more scientific tests to determine how a person learns and what type of learner he is – a listener, a looker or a doer. Two or twenty two questions will not result in an accurate assessment, but Allen's examples at least give you an idea of his learning theory; and as we said earlier, learning is an essential ingredient of memory.

A learner who listens is one who enjoys sounds – especially words – and finds powerful meanings in them. Listeners tend to remember best what they've absorbed through their sense of hearing, rather than from any other sensory perception. Lookers, on the other hand, react best to visual stimuli so anything they see is understood and retained more efficiently. The doers are individuals who like to roll up their sleeves and dig into the trenches.

They put emphasis on practical experience; to them, doing things hands-on holds more meaning.

It is rare for anyone to learn things exclusively in one style. He says the best form of attack would be to combine all three learning styles and adapt each one to a given situation.

Concentration

You can have the best tutor for memory building, but if you can't concentrate, it would be difficult to have much of an efficient memory.

Concentrating is a difficult art to master; look how much technology has taken over our lives. In the mind-training courses he took throughout his life, Allen says there is one technique that might help some individuals develop their concentration skills. This one is adopted from a Far Eastern culture, he says, and is a century-old practice, but is still valuable. It sounds easy enough but your initial efforts at actually doing it may seem futile:

- ✓ Light a candle and set it on a table where you can see it clearly;
- ✓ Stare at the candle for two minutes and take in every detail: color, wax, size, the flickering of the flame, etc.;
- ✓ Close your eyes and keep the image of the candle in your mind's eye – hold this image for as long as you can;
- ✓ Don't be discouraged by your first or second attempt. Keep trying until you can hold the image of the candle as long as you can.

Now that we've dealt with the two indispensable ingredients for memory building, let's "concentrate" on the ways to improve our memory: What do you think is the most fundamental tenet for improving your memory? Take care of yourself!

Body and mind are one. Don't kid yourself thinking that you can set about your merry way doing things you want to do and neglect your physical self. The following rules, are things you hear repeatedly. They still have their weight in gold – old advice, but good advice. So pay heed to them:

Sufficient sleep

Not having enough sleep is a barrier to concentration and learning. On those days where you had insufficient rest and sleep, were you as

productive at work or in school? Did you remember more and retain more, or did you brain feel fuzzy?

Balanced Meals

The experts have stressed, over and over again, that a good, balanced meal is therapy for stressed-out and burned-out lives. Good, healthy eating is essential for our memory to function at an optimal rate. Without the fuel our body needs, how can we expect our mind to be at its peak performance? Fresh and vegetables should fill your plate. According to researchers, those who eat breakfast have heightened powers of recalls more so than those who skip this important meal of the day.

Fresh Air

To benefit from fresh air, learn to breathe properly, and to be concerned about the air quality in our homes and offices. This means that as much as possible, we should have a window open while working, maintaining a comfortable temperature in the room. Stale air that is not allowed to circulate properly affects our concentration and mental processing abilities .

Physical Exercise

Not many people appreciate aerobics or weight-lifting. If you're one of them, go for long walks or swim laps. The idea is to exercise at least 30 minutes per session most days of the week.

Alcohol and Smoking

Big NO! The famous "hang-over" we talk about after a night of partying and "boozing-up" can impair our thinking, concentration and memory. For our memory to work, eliminate alcohol and smoking from your routine especially if you do it to excess. An occasional slip-up may "produce a mere memory blip, but long-term abuse can mess up your mind in various unpleasant ways. Loss of memory will certainly be one of them."

Chapter 3: Memory and Your Daily Life

In a rapidly changing society like ours, it is necessary to adapt to the dynamism of the society. The untrained memory of a person may find it exhausting to remember all the significant information that has happened in a single day. There's too much to remember; important appointment's, special dates, daily tasks, shopping lists, and a list of people to call every day.

The tasks seem endless, thus the mind has to find a way to adapt to this rapid shift of attention. Some of us are able to multitask, while others suffer the inconvenience of attention deficiency and lack of focus, leading to a less functional and productive being.

Forgetfulness

It goes without saying that it is much simpler to forget than to remember. Usually, we easily forget the name of a new acquaintance, or the date of the day before yesterday, merely because we don't pay enough attention to such information. One of the reasons memory-improvement techniques work is because they impose an extent of attention that is otherwise lacking.

Why do we forget?

Forgetting is not that bad. If we don't forget, our minds will be full of unimportant things. It would be even harder for us to search our memory and recover the important ones. Forgetting the trivial information may help you retain and recover only the important details.

But how exactly do we forget? There may be more complex answers, but one way of looking at it is through the three Rs of memory: recording, retaining and retrieving. (Do not confuse these with the three Rs of memory measure, which are recalling, recognition, and relearning).

Recording, retaining, and retrieving are the stages of memorizing. In order to remember something, you have to succeed in all the three stages. However, to forget a thing, you only have to succeed at any one of them. Therefore, you have three chances to forget and only one chance to forget.

Memory Loss

Memory loss is a condition wherein the subject experiences forgetfulness in names, significant dates, recent conversations, telephone numbers, and things you recently learned. It also includes forgetting why you entered a specific room or losing and misplacing things.

Causes of Memory Loss

Depression , which is commonly related to aging, causes lack of focus and attention that can affect memory. You may not realize it, but reducing depression may improve mood and memory problems.

Alcohol and illicit drug use can cause B1 (thiamine) deficiencies that are harmful to the memory. Both alcohol and illicit drugs contain substances that can change chemicals in the brain, thus affecting memory.

Head injury or any blow in the head can be a major factor in memory loss. Usually, memory losses that are caused by head trauma stay as they are or eventually get better, but they don't get worse.

Stress, usually caused by emotional trauma, causes memory loss. In extremely rare cases, excessive stress can result to a condition called psychogenic amnesia. This condition can drive subjects to wander around without remembering either their own names or their birthdates.

Medications can interfere with our memory. These medications include prescription sleeping pills, over-the-counter antihistamines, anti-anxiety, antidepressants, and medications used for schizophrenia.

Infections like HIV, tuberculosis, herpes, among other infections in any lining or chemical in the brain, may suffer memory failures .

Sleep deprivation, whether from insomnia, sleep apnea, or merely stress, can affect memory. Lack of quality sleep has so much impact in one's body.

Nutritional deficiencies can affect memory. Lack of vitamins B1 and B12 can directly damage your memory functions. However, these can be treated with pills or injected medicine.

Thyroid dysfunction - underactive or hyperactive thyroid – may interfere in recalling recent events.

Normal aging also affects memory. As people grow older, subjects with untrained memory may find it harder to recall the types of information like

names of individuals and certain dates.

Mild cognitive impairment (MCI) is characterized by early or unexpected memory deficit. However, this condition is mild enough not to interfere with the daily life.

Dementia, a condition of increasing memory impairment that is severe enough to affect your daily activities. Dementia has many causes, but among the most common is Alzheimer's disease (AD), a disorder where there is progressive loss of brain cells accom- panied by other brain abnormalities.

Multitasking

You may have endless Sticky Notes on your desktop and Post- it notes on your desk for the things that you have to remember. Well, you are not alone. People in this generation have to find ways of coping to this highly busy society. Most people adapt by multitasking, and if you are rather good at balancing many tasks at a time, you may be considered a “heavy multitasker.”

Multitasking does not necessarily mean doing many things at once. Instead, it is the art of attending, performing, and accomplishing many tasks at a certain time, quickly shifting your attention to one job to another. Conventional knowl- edge teaches us that people who can multitask are more productive than others who can’t.

However, more recent researchers believe multitasking actually impairs productivity. Switching from one task to another makes it even more challenging to avoid distractions, and focus on the task at hand. According to research, multitasking is known to reduce productivity by 40% and can cause mental blocks, slowing down anyone’s productivity.

However, the subject of multitasking still in infancy and many questions on the subject remains unanswered. It is undeniable, though, that multitasking works for less complex tasks.

Absent-Mindedness

Absent-mindedness is a cognitive condition wherein the subjects demonstrate a low level of attention, forgetfulness, frequent distraction, and extreme focus on one subject to the extent of being oblivious to everything

else. This condition is not a clinical problem. However, it can seriously damage the subject’s daily lifestyle when ignored. Furthermore, absent-mindedness can also be symptomatic of various psychological and neurological disorders, such as ADHD (Attention Deficit Hyperactivity Disorder), Alzheimer's disease, and depression.

Among the most noticeable aspects of absent-mindedness is forgetfulness. The absent-minded individual usually forget what they came for in a certain location, where the keys and the pins are, where they were supposed to go, and what tasks they were supposed to perform. In most cases, affected individuals only recall these things after it is too late.

The opposite of absentmindedness is focused attention, the sense of concentration on one source of sensory information. In cases of absent-mindedness, the affected individual is experiencing consequences of divided attention, a condition wherein your brain is attending to more than one source of information, causing impairment to the quality of retaining information. Your “mental effort” or the brain’s limited attention resources, may be impaired once you focus on more than one complicated task. When this capacity is less than what is required for certain task/tasks, failures tend to occur.

Because this capacity is limited, it is necessary to also regulate its demands.

Therefore, individuals will do better when they focus their attention to one task at a time. This is referred to as selective attention. Instead of dividing the attention among many inputs at once, it is best to concentrate on one thing at once to ensure quality storage and retention of information.

Chapter 4: Common Memory Problems

It is normal to forget. Even healthy people miss an event or two at times. The question lies in the frequency and the intensity of these memory failures. The following are examples of common normal memory problems.

Persistence

There are things we simply forget. However, in some cases, some people are haunted by memories they wish they could forget but they can't. Persistence refers to consistent recollection of events and information like negative feelings, traumatic events, and current fears. People who are suffering from depression are more likely subjects of persistence; so are people with PTSD or post-traumatic stress disorder.

PTSD can be a result of various physical, emotional, and psychological traumatic exposures, such as wartime experi- ences and sexual abuse. Some forms of persistence are called flashbacks. Some may portray horrifying events in real life, while some are mere negative distortions of reality.

Transience

Transience may normally occur as people age. This condition is the tendency to forget information over time. Although it seems like a memory weakness, transience is actually regarded as beneficial by brain scientists, since it is the brain from unused, trivial memory, thus making space for more important ones. Memories that seem important and are frequently used are least likely to be forgotten .

Blocking

That moment when you are certain that you know the an- swer to a specific question but you just can't quite say it is called blocking. It is the temporary inability to retrieve memory.

Generally, the hindrance to this inability is a memory similar to what you are looking for. You try so hard to retrieve a memory and you only remember the wrong one. A friend asks you a question and you are certain that you know the answer, but you just can't say it, yet it's on the tip of your tongue. The competing memory in this case is very intrusive, as it hinders you from finding the right one. This is primarily the most common type of

mental block. Blocking is known to occur in all ages, but it becomes more frequent with age.

Misattribution

This condition occurs when you recall bits of accurate information, but misattribute some details such as the place, the time, or the people involved. Another known form of mis- attribution occurs when an idea pops up in your memory and you thought the idea was originally yours, when in fact you have only read or heard it somewhere else. This may explain unintentional plagiarism cases by writers, who mistakenly passed subconsciously copied information. Just like other types of memory lapses, misattribution grows more frequent with age.

Suggestibility

Suggestibility is a condition wherein your memory be- comes vulnerable to suggestions, information that you learn about an event after the memory has been incorporated into your brain. This additional information adds up to your recollection of the event even if you haven’t really experienced it.

Currently, only little is known about how suggestibility functions in the brain. This condition is currently a hot subject in the study of brains .

Bias

Bias is a condition wherein the perceptions stored in your memory are filtered by your personal biases, beliefs, experiences, background, previous knowledge on the subject, and even your mood at the moment. People’s attitudes and preconceived ideas bias their memories, and these biases can influence the quantity and the quality of information you can recall.

Chapter 5: Memory Palaces

You might know who Sherlock Holmes is, and for a good reason. Not only is he the most famous fictional detective there ever was, but his amazing feats of being able to solve countless mysteries and puzzles are all thanks to his powerful use of memory retrieval techniques. He makes use of what is known as the memory palace technique.

The memory palace is one of the most powerful memory techniques available, but what makes it even more interesting is that it's been around since ancient Rome. According to the popular myth, this technique was developed by Simonides, a Greek poet. He was attending a banquet when two men called him outside. When he arrived outside, the building behind him collapsed, crushing everyone who attended the party. However, Simonides was able to identify all the crushed bodies inside, supposedly because he remembered where and who sat at every location in the banquet hall. This technique of being able to remember based on location came to be known as the memory palace or mind palace.

In order to do this, one must first visualize a place or location he or she is familiar with. This can be the layout of a house or the directions to the daily commute to work. This place will be used to store the set of memories used for physically remembering memories.

Choose a place

The effectiveness of this technique is based on how familiar you are with your chosen place. So make sure this is something that you know very well, without making too much effort to make out the details of the particular place.

The next is to ensure that you build a specific walk-through in that place instead of picturing it as a still-life scene. For example, you may visualize the neighborhood in your work area, so you can try to picture out the usual route you take when going to your workplace. If using your own house, mentally walk through the door where you usually enter, all the way through the exit. This will help you keep track of items you will be recalling in order later on.

List the distinctive features

As you start your mental walk-through, take note of the things and distinctive features you encounter. For example, you might notice your bright blue front door when you enter the house. As you go inside, you are standing right next to a yellow coat hanger. Then moving along, you pass by the TV room with the big potted plant near the doorway. As you move along, continue to note these features and make them stand out. These are the details that will be of utmost importance to you later on.

Have the place imprinted on your mind

This step is crucial because the ability to remember and make a memory of the things you want to recall relies on how well you know your memory palace. This is why it is recommended that you choose a place or a house that you know very well. To make sure that you are 100% familiar with it, the following methods can be performed:

- ✓ Do a physical walk-through inside the house, noting the distinctive features you want to employ later on. Repeat these things out loud for retention.
- ✓ Write down these distinctive features and repeat them out loud while mentally walking through them.
- ✓ Ensure that every time you do this mental walk-through, you are seeing these distinctive features from the same vantage point.

What you need to realize is that you can never get too familiar with this memory palace of yours. In fact, it is better that you “overlearn” this so that it becomes muscle memory the more you practice recalling this .

Designate and associate

Now that you have your visual associations in place, you can take each distinctive feature and associate it with the element you want to memorize. In fact, there is a right way to do this. To ensure that retention happens, make the association as crazy and as absurd as possible. Make it offensive, animated, unusual, and nonsensical – all the things that are sure to pop and stand out in your head. Make it so unique that the possibility of it happening in real life is zero.

Let’s go back to that bright blue door. Now, imagine you are using your memory palace to remember your grocery list. You may picture out the

bright blue door with a frame of a single egg hanging. This is, in fact, Humpty Dumpty and he's about to fall off the hanging frame. Next, you go to your yellow coat stand. There is an army of broccolis – little green veggies that are trying to climb up the stand. As you can see, your grocery list is hidden among the few details peppered throughout the house, coupled with an absurd animation and story. Do the same for the rest of the memory palace and make the story as crazy as you can.

Retrieval and recall

Now that you've built your memory palace, you can revisit this place anytime you want in order to recall the associated items. You will soon discover that with practice in visualizing, you will be able to effortlessly retrieve the memorized items instantly as you look at the journey's selected features. Walk through the memory palace a couple more times, and when you find yourself at the end of the journey, imagine turning back around in the opposite direction and back to the start.

As humans, we tend to be generally bad at remembering things. But the memory palace is all about changing your memories into images that are more familiar and to place them in a mental location that you know very well. Since information and data tend to be just floating around in our brains, our goal is to give these things a structure for them to anchor on. When that happens, we provide context and order to the things we want to remember .

While the use of the memory palace is still relatively unknown, it can be very helpful in aiding you to memorize large bits of information. This technique can be used by a student trying to pass an exam, or a traveler who wants to be fluent in Spanish, or even to an ordinary person who just wants to stick to the grocery list on a run to the mall. Whatever the desired outcome is, the memory palace is a great way to help you achieve your goals.

Perhaps one of the most important benefits one gets from the memory palace technique is its use for our ageing minds. We're all ageing as we speak and even as you are reading this book. With age comes deterioration, and our memories get weaker. There are certain situations that can leave us feeling frustrated due to these changes. But studies have shown that the use of memory palace holds promise in aiding the enhancement of the ageing

brain. While it doesn’t reverse the weakening of the memories, it can be the perfect solution towards occasional forgetfulness. And with constant practice, the brain becomes more elastic.

While the basic idea of creating a memory palace may sound all too simple, it can actually take you a couple of tries and constant practice to get the hang of it. The concept sounds basic: choose a place that you know very well, imagine yourself walking around this place of your choosing, and place objects that remind you of the items as you move around. Easy right? But don’t be surprised if you find yourself screwing it up on your first few tries.

The first and most common mistake is that you may be merely imagining yourself placing objects on this memory palace. Most people do this by imagining themselves placing bread on a table, for example. And then later on, they get frustrated because while they remember putting bread on the table, they forget what or where it was. The problem with just imagining yourself placing an object in a scene is that there is nothing that connects you to it emotionally. Moreover, random items are much harder to remember than deliberate things that share an emotional connection with you. To increase emotional responses, use people especially ones that you personally know or who are popular. You will get a better result if you choose easily recognizable person, such as those who you find appealing, or those you look up to, as these can already elicit an emotional response to you. However, try to mix it up. If you choose people who are all models that you admire, for example, you will most likely mix them all up. So try to give some variation, let’s say a model, a singer, a president, and an athlete you all look up to.

When choosing the place for your memory palace, you have to consider the size in relation to the number of items you need to recall. Let’s say you only need to remember 60 items, so you pick out a small apartment you once lived in, or your dormitory room. But if you’re planning to remember a whole lot of things, then choose a bigger memory palace, let’s say your entire high school or the layout of the city block you know very well. If you can’t decide just yet, you can use the home you currently live in as default. You don’t need to overthink this part, as long as you are very familiar with the place you chose for the memory palace.

As you are getting started with your memory palace building, imagine yourself walking, touching things, seeing details, noticing all the colors, smelling odours, and even feeling emotional reactions to the objects surrounding your memory palace. The more details you add to the scenes, the better you will remember it. There is no harm in overdoing this step, as you will start to have a notion of how much you need to focus on each image as you progress, as well as the amount on whether the things on the list are temporary or permanent. A memory palace is extremely personal. Normally, many of the associations that are made would make sense only to you and to you alone. Do not stress about not finding a lot of different associations at first. Instead, put emphasis and focus on sensory perception, especially the visual detail of images invoked. Pay attention to the smells, to how the images interact with the environment, and so on. Remember that your palace is the location and the images come and go depending on the things you are memorizing, so make sure these all interact with each other. That reason alone will make you memorize more items on the list, if for no other reason because you will spend more time thinking about them. There are no hard rules on how to create images that can easily be conjured by you at will. The way you learn will be one that's trial and error – all about finding the ones that work for you and what doesn't. Most of the time, normal memory will fill in the gaps if you are familiar with the item and if you access the information quite relatively soon after storing it, which is the case with recalling temporary lists. It is also often helpful to think of the scenes in a three-dimensional manner, even if you are trying to remember a drawing or another two-dimensional image.

Another thing you need to keep in mind when creating memory palaces is to avoid overcrowding. Try to associate only one image to one part of the Palace or an object in it so that you can always visualize every image individually. Additionally, follow some sort of basic order so you can always remember what comes next in each palace you create. For a small list, you can go from a simple entrance in the house to the exit – just like that. But for longer lists, you may want to go through the front door, inside the living room, then the couch, out to the balcony, back into the kitchen, then to the wall clock above the refrigerator, up the stairs, down the hallway, through the bookshelves, inside your room, out the window, to the second balcony then finally taking the stairs back down to the front lawn and out the house. This is just an example, so you can use as much detailed

strategic locations in the place you are familiar with that you will be using for your memory palace. The greater effort you invest in the groundwork, the better the results will be.

To give your memory palace a spin, try a small list to begin with. A set of 10 items will be sufficient. Make sure you count each recalled item out loud before writing it down, taking note of how long it took you to regurgitate the whole list. By counting out loud, you “reset” your working memory and will be able to make a fresh start. An example of a 10-item list is: cinnamon, asparagus, ketchup, salmon, deodorant, steak, ice cream, pizza, toothpaste, salad dressing. Use these set of items as the first exercise. Results will vary, but practice makes perfect. Do not be discouraged if you cannot recall him the first-time round.

Chapter 6: The Peg System

The Peg system was also referred to as the link system in the ancient times. This method involves linking numbers, items, letters, or images to another object, thus creating a peg. The Peg system is useful when memorizing short lists. Ideally, the peg you have created can be used more than once. It is recommended, so you don't have to create a new peg for every new list. I will discuss more about the system later in the succeeding chapters and give more sufficient examples.

This system works well with the Memory Palace. It is commonly used by world record holders in memory recall. In this system, items to be remembered are pegged to or associated with certain images in a certain order. The idea for this system can be traced back to the 1600s, when it was developed by a certain Henry Herdson. He linked a number with any objects that resembled it.

The peg system uses peg words to act as mental pegs from which you can hang the information you need to remember. For example, "candle" is for number one since it resembles the form. One of the great advantages of this system over the memory palace or the method of loci is that you don't have to be dependent on retrieving items in a fixed sequence. You can virtually access any item on your list without having to work your way through systematically. However, it is a bit more complicated to remember since you will need to make a lot of associations and know these inside out.

One of the best ways to explain the complex system is by envisioning a deck of cards. There are 52 cards in the deck, with 13 of each suit. A peg system which would be used to memorize the order of this deck would only require 17 images. In this simplest application of the peg system, the images would be divided into the following: one for each suit and one for the rank of the given card. This means there would be an image for perhaps the "king", and one for the "ace", and one of each of the rest. So this gives us 13 images in total to work with. But even then, this would not be enough to memorize all the suits and the ranks. This is where you will need another four images, one for spades, one for diamonds, one for hearts, and one for clubs. Each card will need to be represented or associated with two images. Let's take the ace of clubs as an example. Let's say your image for the ace

is an athlete – it could be your favorite baseball player or your favorite football player. The image you assign to a club is a trophy. So if you have the ace of clubs as the first card in the deck, you may think of your favorite athlete holding the trophy in a triumphant blaze of glory. The more graphic the action is, the better. You will now be entering your memory palace. Head to the first landmark and hang the first peg there to help you make the association for recall. Repeat the process with the other landmarks we covered early and you have yourself an imaginary narrative. As you progress in your training, you can start to associate one image with one card. 52 might seem more daunting than 17, but it will stick more permanently.

We have already discussed about the PAO (person – action – object) system and how it is one of the most effective yet complex peg systems available. The peg system can also be used in other ways. Some chess masters use it to make mnemonics for chess competitions, and other use it for speedcubing, which is the practice of solving a Rubik's cube in the least amount of time you can. In speedcubing, you would require many complex algorithms with many moves each as opposed to basic solutions that only require a dozen or so algorithms. The peg system also has another advantage - it facilitates long-term memorization. This is because the process slowly converts the codified images into decodified memory. By using the peg system, you can apply mnemonics in mathematical endeavors – for example, memorizing the sequence of numbers in the number pi. It may sound funny and nearly impossible to address mnemonics to a whole set of numbers, but it works.

Before starting, it helps if you have previous experience with using the Memory Palace. Together with the peg system, memorizing 20 digits under 2 minutes is a feat that becomes possible, even if it seems too overwhelming to try. If you have tried to use the list of 10 items in the previous paragraph, then you will should have sufficient practice. You will be using a few famous individuals to function as a connection between the location in your memory and the item you wanted to memorize. This should improve your recall time considerably as well.

The most common ways with implementing the peg system are the Rhyme System and the Dominic System. In the Rhyme System, words are chosen that rhyme with the number they are associated with, and refer to a

real or tangible person or object. A common example is associating the word “one” with the sound of “gun”. You can also take it a step further and apply the PAO method, which is associating person-action-object. You can set it as something that rolls off your tongue like a nursery rhyme, such as: “Billy the Kid shooting a gun.” Repeat this process for numbers 2-10 if you have a 10-number sequence. There is a downside to using rhymes though, so be warned. As you progress and extend your sequence, numbers start to appear with repeat endings. To give you an idea of how numbers can be associated with rhyming words, take a look at this example.

1 – bun, 2 – shoe, 3 – tree, 4 –door, 5 – hive, 6 – sticks, 7 – heaven, 8 – gate, 9 – vine, 10 – hen.

As you may notice, these rhymes come from a popular nursery rhyme that makes it quite easy to remember. However, if you are not familiar with the rhyme, it may not evoke a strong emotional response. In which case, use something else that resonates with you.

Another option is through the use of the Dominic System. Dominic O’Brien, who was the creator of this system, was an 8-time World Memory Champion. His system promises an easy yet flexible method for creating pegs. You associate one letter for each digit from 0-9, then create a peg for a two-digit number by reading the two letters as the initials of a person of your peg. The associations in the Dominic system are as follows (digit to mnemonic): 1- A, 2 – B, 3 – C, 4 – D, 5 – E, 6 – F, 7 – G, 8 – H, 9 – N, 0 – O. This should be easy to recall as it follows the course of a normal mnemonic system. Suppose you are creating a peg for number “15” using the Dominic system. “1” is associated with “A” and “5” is associated with “E”. SO you will end up with the mnemonic “AE”. Now think of a famous person who has the initials “AE”. You can use Albert Einstein as an example. To strengthen this association, you can add in an action like “Albert Einstein writing on a piece of paper”. You now have your peg for the number “15”. Now test yourself by memorizing a hundred digits (1-100 in order is recommended, depending on whether you are a challenge junkie). Somewhere along the list, there is a “15”. Once you can remember this peg you have set for yourself, you can slot it in its respective place. With enough repetition, your pegs will simply come to mind with little or no effort at all. It will be like reading a word and understanding what it means even without thinking too much about it. You can follow this same

principle with the rest of the two-digit numbers, or you can take it a step further by combining 3. Just make sure you are able to memorize all your pegs by heart. You can use this technique to recall phone numbers at first. As you progress and as your process becomes more streamlined, you can move on to memorizing large series of digits. You may even end up recalling all the numbers in the value of pi by heart, if that tickles your fancy. In the Dominic system, you are free to play with the definition of the abbreviations you set, but make sure these associations are consistent. If you forget who “OG” or “LGD” is because you were tinkering with different associations, then your use of the peg system isn’t going to help you much going forward. While these rules aren’t set in stone, they are useful as guidelines.

A considerably older peg system is more commonly used by those in the senior demographic. It is also more complex than the first two systems we have discussed. This classic is called the Major System. Essentially, it contains some of the same principles as the Dominic System, by associating numbers with letters. Where this differs is the Major system allows you to create any word instead of only initials, even whole sentences. Every word correlates with a single number in this system, but that same number can correlate with many words. With the application of the Major system, you can determine the number of the given word by the letters that make it. Letters that sound the same are grouped together. For example, 0 = Zero, with a Z. It also sounds like S as in Sero. Or X as in Xero. So 0 = S, X, Z. The creativity is extended to similarities, like downstrokes. 1 is be assigned to t and d, as they both have one downstroke. When the strokes extend, so do the numbers like 2=n and 3=m. 4 is assigned to r because it is spelt fouR. We thus have the associations below:

0 = s,x,z , 1 = t,d , 2=n , 3=m , 4=r , 5=l , 6=sh,ch,j,soft g , 7=c,k,hard g, q , 8=f,v , 9=p, b

The immediate difference that you will notice is that the Major system uses more words and letters than the Dominic system does. However, it does not use all of them. You will also notice that there is an absence of every vowel, along with the letters “h” and “w”, which are unassigned letters. They can be used at any place without affecting the “number” of the word because they essentially have no value. So how does one determine the correspondent number? This is done by the assigned letters, in order.

For example, if a letter in in a word is assigned, it should be replaced by the correspondent number. So if a word's first letter is "f", then the first digit is an 8. But if you are going from a number to word, which is the opposite of the first step below, you can use the unassigned letters freely, and if a digit has more than one assigned letter, you get to choose which one to use among them. The most important thing is that you get to the end result of using an actual word with which you are already familiar with.

To do this, we will look at the example of the word "Titan". To discover its number, remove first the unassigned letters, and in this case, these are "A" and "I". so that will leave you with the "TTN" as a word. Next, convert the letters to their equivalent numbers, so that's 1 for "T" and 2 for "N". Following these rules, you will end up with the value 112 for the word "Titan".

If you wish to do the reverse, which is going from a number to a word, the technique is more or less the same. When you see "112", you will immediately think of the possibilities: that 1 can be "T" or "D", or that 2 is "N". With this, you can have a number of possible words such as "Edition" (eDiTioN), or "duty on" (DuTy oN) – but spaces should be ignored, so this can be wiped out from the list of possibilities. This is especially true if "112" was just a part of a larger string of digits and "duty on" was more adequate to form a sentence. You can create a peg so that every 2-digit number, complete with an appropriate person-action-object (PAO) technique, becomes even more interesting. By this stage, you will realize that pegs are considerably more demanding than the Memory Palace. However, the pay-off has a massive impact on your ability to recall lists or a series of numbers. The gap between 10 and 100 in which you will assign pegs to each number sandwiched inside will be the largest investment you will ever have to make. But the rewards don't come immediately, as it not easy to learn and master. It will understandably take time to see the results on your end. The solid use of pegs can be used to improve your daily life. Numbers after all, are very much prevalent in our lives. They can be in social security numbers, passwords, and a whole lot more. As far as creating pegs go, the Major system allows for a lot of imaginative possibilities. You can make it as personal as you want.

There are other peg systems you can use, and it can be entirely relative to your own understanding of how digits work and what they mean to you.

For example, you can assign the word “me” to “1”. This makes sense because there is only one of you. Or you can assign a pitchfork with three prongs to be your associated peg for the number 3. Or a simple association of the “hand” to mean 5 (representing the five fingers) can also work.

Now to put it to practice. Make yourself a list and create ten pegs for each of the digits from 0 to 9. Choose whatever method works for you. Also, make sure you write down your peg list so you can have something to look back on as a reference guide. Next, try to get a randomized series of numbers. You can generate this by yourself manually or use an app on the phone. Any software on the computer will work just as well. Start with 10 or 20 digits to begin with. If you clear this hurdle. progress to 50, and then 100. With your peg list in hand, try and memorize the digits using a combination of the Peg system or any technique of your choice, along with the Memory Palace. Place your pegs at your chosen checkpoints inside the Palace. Everything else is entirely up to you. Initially you will falter and that is part of the process, especially since your brain is still adjusting itself to a new system. Don’t fret if you cannot do all 50 at once or if you get a few wrong. This is bound to happen at the beginning and is completely normal. The important thing is to realize how much easier it is to recall a person or object doing an action, than it is to remember a random number. Take baby steps in your own time. Do not start with 100 pegs before you even have the first 10 nailed down. With practice, this will become second nature.

All these memory enhancing techniques can be used for everyday life. While exercises such as the peg system can be more helpful in competitive play, there is certainly no limitation. You can use it to remember your grocery list, phone numbers, house addresses, and any relevant information. Another advantage is that you will be subjecting your brain to rigorous training, and in turn, keep yourself sharp. With brain training, individuals can become more efficient at performing everyday tasks of varying complexity. Think of it as gym time for your brain – mental success is key to long-term success and health.

Chapter 7: The Mind’s Eye

There is a saying that goes, “the eyes look, but the brain sees.” Is there really a difference between looking and seeing?

We think of the eyes as cameras, able to capture the images that we see and interact with in the outside world. In reality, our view of the world is a result of the illusion our individual brains process from what our eyes input into it. This input is also constantly shifting, and our brains tend to filter out unnecessary movements as it scans the environment in detail. From this information, the brain translates it into an image that we can understand. In fact, the brain is only able to receive three images for each second, and these are designated and associated with earlier memories or information in order to provide the experience of your current reality. Think of it as a machine that runs constantly at all times. In order to protect the brain from overheating, it needs to save energy by choosing what is worth looking at.

In this regard, we may be familiar with the feeling of looking at something inside our brains, conjuring up images even when these images aren’t directly in front of us. This is what we call the mind’s eye. It's the mental ability of being able to conceive imaginary or recollected scenes. It is akin to the mental picture we get in our mind. If I tell you to think of a pink elephant on tiptoes, balancing on a ball, you would be able to conjure this image just as easily as if you’ve seen it in real life. In brain scans, neural patterns light up when you are given the task to imagine something with your mind’s eye. This indicates that there is a part of your brain that actively seeks to come up with the image or recollected images.

As a memory enhancing technique, mind’s eye refers to a method known as sensory discrimination. It requires you to ignore distractions when we are actively focused on something we are paying attention to. In the Mind’s eye exercise, you can be asked to remember a target image and point out whether a set of images contain that specific target image. It targets visual memory and retention as the images become more similar, changes patterns, or if you are tasked to remember more than one image at a time.

The use of the mind’s eye is undoubtedly an important aspect of building your memory palace. How else will you be able to visualize the house you will be placing your associations on? And if you put animations on your

associations, you will need to create vivid images in order to make them memorable. This form of visualization needs to be sharpened up in order to enhance recall skills and mental imagery. Think of memories as encoded like a jigsaw puzzle, with associations linked together. The stronger and more vivid these associations are, the more effective one can access a memory.

Let's take a look at how the famous detective, Sherlock Holmes, does this technique. In "A Scandal in Bohemia," there is an exchange between Holmes and his partner, Watson, that goes like this:

"Quite so," he answered, lighting a cigarette, and throwing himself down into an armchair. "You see, but you do not observe. The distinction is clear. For example, you have frequently seen the steps that lead up from the hall to this room."

"Frequently."

"How often?"

"Well, some hundreds of times."

"Then how many are there?"

"How many? I don't know."

"Quite so! You have not observed. And yet you have seen. That is just my point. Now, I know that there are seventeen steps because I have both seen and observed."

This exchange shows us that Holmes has developed keen observational skills regardless of any environment he was in. He was never observing, and never out of touch with his surroundings. This kind of mindfulness is almost superhuman, yet with constant practice, this feat can be performed by anyone just as easily.

We often block our sense to the world, using only one or two sensations, when, in fact, we can experience situations through multi-sensory perceptions. This means we should limit ourselves to our senses – use your sense of hearing, touch, taste, or smell when appropriate. While Holmes' keen observation astounds us, it is certainly not a superhuman ability. He wasn't born with it, but he spent a lifetime cultivating habits of mindfulness. He chose to observe, at all times, regardless of circumstances.

Our worst habit is not paying enough attention. This is even more pronounced in this day and age, wherein we would always choose to distract and block ourselves from the outside world by putting on headphones, or wearing dark sunglasses. We choose to limit the use of our senses, thereby allowing us to not fully enjoy and bask in the details of our surroundings. But despite these bad habits, the great thing about our brains is that we can always rewire the way it works, in a way that mindfulness eventually becomes less of an effort.

Work into it every day

Putting in the effort will pay off, and because you need observational skills to become a habit, you will need to do it every single day. You can try to take in one interesting picture a day, or take note of your co-worker's tie daily. Whatever it is, choose to become more observant about one specific thing until it starts to become effortless.

Write it down

One of the best and effective ways to help you remember and become observant is to write or draw the things you observe. It can be done at any time of the day, regardless of what you are doing. For example, dedicate 5 minutes during your break time at work and write down the things you observe from your colleague's behaviour. Pay attention to how they talk, how they animate when talking, their gestures, or how they take a sip of water. The more you do this on paper, the better you get at observing .

Start making connections

Just like using the memory palace, associations are important. After all, the increased focus and critical thinking aren't going to help you much when you can't make connections between the knowledge you have and what you see.

Expand your knowledge base

Remain curious about a lot of different things. This will help you increase your associations and make them even more interesting and relatable to you.

The memory palace and mind's eye are concepts that not only improve memory, but can also be used as coping strategies for people who have

difficulty with visualization. In the case of children with Attention-deficit Hyperactivity Disorder, having a weak working memory is something that is all too common. One way to go about this is assigning objects that rhyme with numbers. Let’s say, 1-sun, two-shoe, 3-tree, and so on. In order to learn new information, the child can simply visualize this number rhyme interacting with the things to memorize. If he is tasked to bring a book, an eraser, and a notebook, he might do it this way: a book about suns and planets, a shoe with an eraser on the bottom so it doesn’t ever leave shoeprints, and a notebook caught high up in a tree. You will soon find out that the ADHD brain loves variety and novelty. Working with your child to improve their weak working memory may take time, but it will be worth it.

Being mindful creates greater awareness between the environment and ourselves. The realm of observing goes beyond just seeing – but involves a closer look at details in order to enhance our memory keeping.

Chapter 8: Method of Loci

“I consider that a man’s brain originally is like a little empty attic, and you have to stock it with such furniture as you choose.”

- Sherlock Holmes

This mnemonic memory technique was by far the most common among the ancient citizens of Rome and Greece. Basically, what ancient Romans used to do was associate information with a considerably familiar place or setting in their lives. The word “loci” means location. People in ancient Rome used to associate information to the places in their neighborhood or part of their houses. Even today, this can be used for people who have trouble remembering things.

In effect, the places will serve as your clues to recollection. This concept may already seem familiar to you by now, especially after the first chapter of this book, but in essence, the method of loci is the basic underlying principle for memory palace. Method of loci is all about remembering something based on a familiar location and its association to an object, while the memory palace or mind palace deals with a pre-learned set of locations that make up a whole “house” or “palace.” In short, memory palace is a sequence of locations that one can find useful for the method of loci.

Apart from using the mind’s eye, the method of loci can also be learned through other senses. Those who are deficient in one sense tend to overcompensate and excel in their remaining senses. But these senses can be trained and developed regardless of whether you have or don’t have the loss of a particular sense. Examples are learning Braille for tactile sense, wine aroma kits for olfactory sense, and perfect pitch exercises for the auditory sense .

Wine educators and their students use aroma kits to increase their knowledge about the specific ingredients found in wine. With constant practice, one can learn to develop a personal scent memory to eventually distinguish the nuances in wines or whisky.

Perfect pitch is the rare ability to be able to identify, sing, or play back the exact same note. If you were to play a C# on the piano, the one with

perfect pitch would be able to tell you that you've just played a C#. And if that piano was out of tune, he'd be the first to know. While most studies show that having perfect pitch is genetic, it is a skill that can be learned early in life. However, as we age, our brains lose some of its neuroplasticity, and it becomes harder for us to train and develop skills like this. Studies claimed that people with perfect pitch can subconsciously allocate musical notes to their memory and will be able to process them later. This gives them the ability to accurately identify notes without any outside reference.

Blind people are known to have far superior perceptual abilities in their remaining senses. One of these is their tactile sense, as evidenced by the mastery of the Braille reading system. However, sighted people can also be trained in simple sensory tasks that can eventually progress later on.

Combining Method of Loci With Memory Palaces

For example, you are attempting to memorize all the presidents of the United States. You can create a "palace" in your mind, or simply picture the structure of your own house or neighborhood. Generally, you only need a very familiar setting pictured in your mind in order to associate each event and setting to a specific president. That way, when you go to a certain room or event in your narrative, the associated president would readily come up.

Imagine setting the things you need to recall along a course you know well or in particular areas in a natural room or building. For example, for a shopping rundown, envision bananas in the portal to your home, a puddle of milk on the couch, eggs going up the stairs, and bread on your bed.

Designing Mnemonics, Imagination, Association, and Location

Imagination, association, and location are the three basic principles of mnemonics. With this guideline, you can generate powerful mnemonic systems by working all three together .

Imagination

For a fast later recall, imagine and visualize the situation more strongly. It will stick in your mind very effectively .

Association

The best way of remembering things is by linking a thing with the things you want to remember. This association can be created:

- ✓ If you place things on top of each other.
- ✓ Rolling things together.
- ✓ Assimilate images together.
- ✓ Covering the things around each other.
- ✓ Make associations by using the same smell, color, shape, or feeling.

Location

With the help of location you can place information into rational situations so it drapes together. It is a way of separating mnemonics from each other.

Transferring Memory Palaces

Now that you've settled on your memory palace, you may begin to wonder: can I transfer from one place to another? If not, can I override, delete, or re-use the same memory palace over and over again?

The quick answer to all these is: yes, you can do all these things. Since our individuality allows us to choose what works best for us, there is simply no limit as to how many times you can recycle the same memory palace over and over again, or to retain more than 10 memory palaces at a time. However, take note that this can be most effective when using long-term memories, and not for high-speed working memories, such as those used in rapid-fire competitive play.

Before you create another memory palace, make sure you are 100% familiar with your current one. The same principle applies when you choose to override the old data in order to make space for new ones. Another technique is to link new data with the old ones, making relevant associations. For example, your first memory palace lists down the presidents of the USA. Then you decide to fill it with a new list of human-crewed spaceships in order. You can associate, say, George Washington riding the Vostok 1, representing the first president of the USA and the first human-crewed spaceship. Do this for the rest of the list until it is fully committed to your memory.

In the case of transferring from one mind palace to another, try to limit the different palaces to less than 6 at a time until you allow yourself to really

immerse those details so deep that it becomes ingrained in you. You can then jump from one palace to another, trying out the many different ways and lists you can train to recall.

Chapter 9: Mnemonic Tricks

Mnemonics is a type of memory tool that helps us learn large pieces of information that takes the form of music, acronym, expression or word, model, ode or rhyme, or connection. The concept behind it is to encode hard-to-remember information and hide it in a way that is much easier to recall afterwards.

When assigning mnemonics, there are three fundamental principles you need to know. These are imagination, association, and location.

Imagination is what you need to create strong and vivid images that are potent enough for you to recall later on. The more potent your imagination and visualization is, the more it will stick with you. This is why it is recommended that you go crazy in making up these mental images. Make it as wild, absurd, violent, or as sensual as you want.

In association, you create linkages between the images and the way you want to remember them.

The use of location is particularly important because it gives you a specific context in which you can recall your information.

Mnemonic Devices

Mnemonic Devices are the tools or strategies a person uses to effectively remember and improve brain function.

Introduced and defended by Gerald Miller in 1967, he contended that the effectiveness of mnemonics to memory improvement is very effective, and his study even showed that his students who are the subject or his study, were able to increase their test scores to as high as 77% when using a mnemonic device. There are basically several types of mnemonic tools.

Here, some types of mnemonics will be given focus, all of which are known and considered as the most common type of all.

Connection Mnemonics

This type of mnemonic deals with information to be remembered that is already known, or associating the items to something that sounds like it. When remembering the direction of latitude and longitude, for example,

you may picture out long lines across the globe, from North to South that coincides with LONGitude. On the other hand, latitude must run from West to East because there is no N in the word, and it rhymes with the word flat, just as flat line runs horizontally.

Clustering

Clustering refers to a series of information grouped together in order to memorize the data more efficiently. In memorizing a telephone number for example; people tend to group the numbers together in three to four digits so it will be easier to remember them. One example is a long number; 1,8008,808,008. Just by looking at it blandly and plain like that, it will be impossible and a little difficult for someone to remember it. However, by clustering or grouping them to remember; 1-800-880-8008, it becomes easier to process and therefore, memorize it.

Name Mnemonics

Name Mnemonics is almost the same as acronyms. In here, you rearrange the first letter of the words so you can form a name of a person or anything so it will be easier for you to recollect and remember. One perfect example is ROY G. BIV. This may look like a name, although this actually stands for the seven colors you would find in the spectrum or rainbow. The name in fact, actually stands for Red, Orange, Yellow, Green, Blue, Indigo, and Violet. People often use this to easily remember a particular list.

Acronym Mnemonics

Taking the first letter of every item you want to remember and arranging them in a way that makes more sense and comprehensible to you is called an acronym. One of the most famous acronyms is ROY G BIV, in which each letter represents the first letter of a colour associated with it: red, orange, yellow, green, blue, indigo, and violet.

Word Mnemonic

This is the most popular form of mnemonic use, and it involves making a phrase or sentence out of the first letter of the items. Similar to acronyms, but retention can be stronger because of the coherence of the made-up sentence. For example, the order of operations for math is Parentheses, Exponents, Multiply, Divide, Add, and Subtract. Taking the first letter of each word will give us PEMDAS, which doesn’t make sense or is hard to

remember. But if we form it into a sentence, such as “Please Excuse My Dear Aunt Sally,” the recall can be very strong.

This sentence actually stands for the order of mathematical operations you must conduct when doing calculations.

The sentence actually stands for Parenthesis, Exponents, Multiply, Divide, Add, and Subtract. You would see that the first letters of the mathematical operations are the same as the first letters of the sentence.

Music Mnemonics

Have you ever wondered why it’s easy to remember the lyric to a song, particularly a song that you really like? The same method of being able to memorize the lyrics of a song can be used to remember things in academics or in daily life. In the same way, advertisers make use of jingles about their products in order for customers to remember them. Tunes are very easy to remember, and if you place the information through a song, you may be able to recall more than you expected. An example of this is how we memorize letters by singing the Alphabet Song.

There are people which easily remember information with the music he or she associates it with. This could be attributed to the music he was listening to at the time he learned the information; or perhaps he made his own rendition of a song by putting the information in as lyrics.

Music mnemonics is also the target of TV Advertisements .

They use rather attractive and repetitive jingles with lyrics that are easy to remember so people immediately remember their products when they hear the song or perhaps remember the song when they see the product. The repetitive lyrics in the jingle also help the shoppers remember the product.

Model Mnemonic

Model mnemonics make use of representations, such as the pyramid model of stages, a circular sequence model, a pie-chart, or a 5-box sequence. This can be seen in samples, such as Maslow’s Hierarchy of Needs, wherein each stage in the pyramid represents a specific need, building its way up towards the top, which is self-actualization.

Another perfect example is the Krebs cycle, or that biochemical process in the body referring to the creation of ATP or energy that a person uses to

fully function. Unless deeply interested in Biology, one would find the written process rather confusing, deep, and thoroughly scientific. By presenting the process through a model however, on will be able to see the smooth flow of the process and successfully memorize the cycle in an efficient manner.

Ode or Rhyme Mnemonics

Rhyme Mnemonics is strongly similar to the music mnemonics. In this case, the information is transferred as a poem. There are also instances wherein words or phrases with similar sounds are used to effectively remember the information. Distinguishing the difference between Cyanate and Cyanide for example, could be used with rhyme mnemonics. In this case, Cyanide is a fatal chemical, and could therefore use “I ate and I died.” “I died” rhymes with cyanide, and perfectly describes the unfortunate after-effect of cyanide as a fatal chemical which causes death.

Find several relevant rhyming words to help you recall. For example, memorizing which months of a year have 30, 31, or 28 days respectively can be found in this rhyme:

“30 days has September

April, June, and Novembe r

All the rest have 31

Except February alone

For it has 28 days clear

And 29 in each Leap Year”

Image Mnemonics

Image mnemonics appeals to your sensory memory, particularly to your photographic memory.

Image mnemonics uses the power of images and pictures associated with the information in order to help you remember it. Images do not have to be perfect, for what’s more important is that you understand it and remember the information it signifies. BAT for an example, refers to a list of depressant drugs; Barbiturates, Alcohol, and Tranquilizers.

One way to remember it and its purpose is to depict it as a rather depressed and gloomy looking bat. The bat in itself is already the information you need, and the bat's state of depression may refer to the purpose or definition of BAT. People often contend that the sillier the image is, the more interesting and easier it is for a person to remember.

There are many other Mnemonic tools you can use, including Note Organization Mnemonics, Outlines, Spelling Mnemonics, and many others. All of these may be used to your convenience. However, keep in mind that the purpose of mnemonics is to easily remember and memorize any particular information. Therefore you should avoid creating fancy and complicated strategies for it will only defeat the purpose of remembering. Instead, use simple and easy to remember notes, items, music or abbreviations for you to remember.

Chapter 10: Mind Maps

Mind mapping is a useful technique that offers a simplified diagrammatic overview of a subject. The mind map was invented by Tony Buzan in 1960s. It utilizes the left and right hemispheres of the brain by working simultaneously, cooperating with each other. The logical, analytical left hemisphere understands and assesses the information, while the intuitive, creative right hemisphere finds a visual form in which to present it. The following is a summary of the various processes associated with each hemisphere; this is to help you understand how mind mapping works.

Left Hemisphere	Right Hemisphere
Speech	Creativity
Analysis	Color perception
Sequencing	Spatial awareness
Logic	Creating an overview
Linear thinking	Day-dreaming
Rational thinking	Intuition
Numbers and word recognition	Face and object recognition

To some extent, we are visual learners, and we tend to learn best with visually stimulating aids. One of the best ways to practice this visual memory learning is through the use of mind mapping. Mind maps make use of organizational structures that include the use of lines, symbols, words, images, or colour. It converts an otherwise monotonous list of information into a highly organized diagram in a logical and relatable way .

It also helps that information in mind map usually have a natural relationship with all other concepts one is trying to retain. Think of it like a city, wherein the city center represents the main idea, while the main roads that lead to it are the key concepts in your thought process. The secondary roads represent the secondary thoughts, and so on. Relevant ideas are given symbols or landmarks. You can choose to write down your mind map and study it visually. But with time and practice with this skill, you can effectively whip up a mind map of your own using your mind’s eye.

A mind map also represents the relative importance of different topics. The central themes are ly defined, and all extraneous information is eliminated. You can see the whole picture and the key details all at the same time.

The five essential characteristics of a typical mind map are as follows:

1. Main idea represents the subject you are trying to make a mind map on. Put it at the center. For example, Kinds of Food.
2. Main theme branches from the central image. In the above example, we can put Go, Glow, and Grow as separate branches from the central image of Food.
3. Branches form a key image from the main theme. You may limit this to a few phrases or words. For example, Go foods give energy, Glow foods give healthy skin, Grow foods make one bigger.
4. Associate these branches with images. Examples of Go foods can be riding a bicycle, Glow foods illuminating, and Grow foods sprouting muscles.
5. Connected modal structure brings together all these concepts. Feel free to branch out into smaller concepts, making sure everything is connected as a whole.

Again, we go back to the famous detective and how he would do mind mapping. How would Sherlock Holmes do mind mapping? Since he does a lot of mystery solving and detecting murders and thefts, he needs excellent mind mapping skills to be able to solve them. First, he would segregate the evidence into symbols and icons. Then he would attach pictures to the collected pieces of evidence with the help of branches. With the map right-facing him, he would be able to map out and follow the developments of the case, by joining pieces of evidence and information and extending the branches into further information. With the linkages organized clearly, he can make use of keywords and use these to unearth archived cases in order to find relevant and similar themes. From this, he can start ticking off tasks from each list, always finding relationships and linkages that could all lead into one possible scenario.

As you can see, mind mapping techniques of Sherlock Holmes are what we would ordinarily use if we were to make a mind map ourselves. It just

takes a bit of practice to do so. You may not be a fictional crime-fighter in the 19th century, but the skills for mind maps can help reveal important connections and patterns in the real world that can be beneficial to you. Apart from enabling you to develop deep knowledge about a specific subject, you can use this technique to do simple tasks and manage them properly.

As you start to use this method, you must first decide the topic of your diagram – this can be anything at all; this is your central idea. It can be also very useful to add an image at the center of the map to represent your topic. Start your map in the middle of the page, as it gives you more room for creativity and organization of ideas.

Use thick, colorful branches spanning out from your map. This will allow your brain to be more alert compared to when you use straight, monochrome lines. Add your main ideas as you add branches to your memory illustration.

Here are a few more of the traditional mind mapping guidelines:

1. Start at the center with an image of the topic, using at least three colors.
2. Use images, symbols, codes, and dimensions throughout your mind map.
3. Select few key words to keep things simple.
4. The lines should be connected, starting from the central image. The central lines are thicker, organic and thinner as they radiate out from the center.
5. Make the lines the same length as the words or images they support.
6. Use multiple colors throughout the mind map for visual stimulation and also to encode or group.
7. Develop your own personal style of mind mapping.
8. Use emphasis and show associations in your mind map.

9. Keep the mind map by using radial hierarchy, numerical order, or outlines to embrace your branches.

10. Most important of all, remember to keep things simple and have fun creating your diagram.

Advantages of Mind Mapping

✓ Mind mapping uses the power of pictures and colors in two-dimensional form.

✓ Mind maps make it easy to memorize, recall, and review important information.

✓ Using mostly keywords forces you to focus on what is important, and helps greatly to reduce the information you have to remember and go over.

✓ You can easily organize thoughts with it.

✓ It is easy to formulate an overview.

✓ It is also easy to make connections or to associate between ideas. Simply draw a line with arrow-shaped endpoints between two ideas on different branches.

✓ Mind mapping is fun.

✓ You can always add more to the diagram.

✓ Planning tasks and reorganizing mind maps becomes a breeze if you use a mind mapping computer program .

<u>Good Mind-Mapping Web Apps</u>

MindMeister

MindMeister was built to facilitate collaboration for mind mapping and brainstorming, with an intuitive, easy-to-use interface. You can collaborate in real-time, discuss changes over live chat, exchange ideas, and even view the change history of a mind map.

iMindMap

iMindMap is designed to be visually stimulating, with curved branches, bright colors, and sharp images to help unleash your creativity. You can

take full control over your map with a customizable drawing experience, or let the program create the whole map for you. There are freehand, styling, and text options available so you can create a unique project. It's also simple to brainstorm, refine, plan, and craft an animated presentation.

SpiderScribe

SpiderScribe is an online mind mapping and brain- storming tool that lets you organize your ideas by connecting notes, files, and calendar events in freeform maps. You can create private or public maps and collaborate with others, as your maps are stored in the Cloud and can be accessed anywhere.

XMind

XMind is a useful open source mind mapping tool for clarifying, organizing, and brainstorming concepts, as well as managing complex information. You can gather ideas and craft them in a logical structure to help identify critical information, uncover patterns, relationships, priorities, and trends.

Coggle

With Coggle, you can brainstorm and produce beautiful notes or mind maps to share and collaborate with others. You can drag and drop images onto your diagrams, view the change history to keep track of edits, and customize the colors within each project.

Freemin d

Freemind is an open source mind mapping tool built in Java, and is available for Windows, Mac, and Linux. It has a wide variety of features, including location based mind mapping, collaboration tools, and restore session support.

Text2MindMap

Text2MindMap is a useful app that helps you organize your thoughts using a simple, online tool. All you have to do is type the text into the text area, use the tab key to indent text lines, and click the Draw Mind Map button to see the figure in full with all the information you just entered.

Popplet

Popplet is a mind mapping tool that is perfect for visually curating your ideas and brainstorms. You can visually record your ideas, inspirations and thoughts, as well as upload text, videos, images, and draw on your canvas.

Bubbl.us

Bubbl.us makes online brainstorming and mind mapping simple. To start, type your central topic to start a tree, and then create another idea at the same level or as a sub-level. The Bubbl.us tree will automatically reposition itself to fit any new ideas you have, or you can move the bubbles around manually.

Mind42

Browser-based Mind42 allows you to manage all of your ideas on one platform for a fast and simple mind mapping experience. You can quickly create, manage, and edit any kind of data structure required, for mind maps to explore ideas in a visually meaningful way.

Chapter 11: Other Methods and Strategies

There were already a number of memory-improvement techniques even in the ancient times. In this section, we will be discussing those widely-known strategies.

The Pythagoras Method

Pythagoras, the renowned mathematician also developed a memory-improvement technique that remains useful now. Although only a little knowledge was available on the life of Pythagoras, the book "Iamblichus' Life of Pythagoras" stated that when lying on the bed at night, he always recalled what occurred on that day to retain as much knowledge as possible. It was said that eventually, he could recall more than one day's events at once, indicating an improved ability of his memory to store information even after they had occurred.

Your teachers' instructions to study in advance for an exam may not have been based on the futile hope you'll do better. Maybe by studying a material a week before an exam, and slowly understanding and absorbing every topic required of you, may just be another demonstration of the Pythagorean method.

The Chunking Technique

This technique is probably the most common even in our times. The Chunking technique basically involves dividing a large amount of information into smaller bits for easier memorization. George A. Miller, a Harvard University researcher, first discussed this technique in a study conducted in the 1950s .

A good example of the chunking technique is when memorizing a long string of numbers. For example the number 640,957,304 can be more easily memorized when broken down to 640-957-304.

The Gematria Method

The Gematria method is an ancient Jewish method of memory improvement. Eran Kantz described in his book "*Where Did Noah Park the Ark* ?" how this method first came to use in 1648. In the Gematria method, the number one was assigned to the first letter of the Hebrew alphabet, the

number two to the second letter, the number three to the third and so on. In other circumstances, numbers and letters were paired according to the likeness of their shapes.

Chapter 12: Practical Uses for Memory Enhancement Techniques

Wim Hof Method

Try to take a deep breath and exhale. Repeat.

Before you started consciously breathing, you probably weren't thinking about breathing at all. This is because breathing is controlled by our autonomic nervous system, the one that is also responsible for involuntary muscle movements, such as the beating of our hearts, the blinking of our eyes, temperature regulation, and breathing.

The Wim Hof Method (WHM) is known as a series of breathing exercises and cold immersion techniques to manipulate the body into reaping benefits, such as enhanced sleep, improved health, and better memory function. This method was developed by Wim Hof, also known as "The Iceman," as a means to control the body's inner thermostat and regulate temperature. By learning to control the autonomic nervous system, one can have incredible control over his own physiology through the power of his mind. This is made possible by a method that involves deep breathing and holding one's breath for extended periods that eventually results in your ability to progressively hold breaths for a long time.

To do this, you first take a deep breath, breathing in more air than you need to breathe out, and then do a full expiration afterwards. Next, you hold your breath after this exhalation, then take in one big inhalation as a 'recovery breath" that resets you back. This surge of fresh oxygen rushes in through the cells of your body, allowing it to receive the full effect of the exercise. By inducing hypoxia through breath-holding, one creates a positive stress response by turning on the survival mode that allows one to withstand extreme temperatures. It facilitates survival during lower oxygen environments.

A study of this method in rats showed that it helps in the proliferation of neural stem cells, which help in cognitive function and memory. Since stem cells are only able to survive in a state of hypoxia or when oxygen is low, inducing this state can be very beneficial. Another way in which WHM can benefit the brain is that when there is low oxygen tension in the blood, it

causes more blood vessels to awaken that may not have been used before, since we only use a limited number of blood vessels every day. By increasing the blood flow to them, the brain can be fully alert, and thus be able to focus and retain more memories. In general, it can help increase focus, energy levels, overall productivity, and keeps a calm mind.

Brain Waves and How to Enhance Them

There are five types of brain waves present in all humans, and each serves an important function to our state of wakefulness, play, or even sleep. Each one helps us cope with various situations, and if either one of them is overproduced or underproduced, it can cause problems. When these brain waves are at their optimum level, it can produce benefits that can improve memory and cognitive functioning. There are various ways to increase them and to enhance each brain frequency.

Gamma Waves

These are involved in higher processing tasks, such as learning, memory, and information processing – all higher cognitive functioning around 40Hz gamma waves is thought to be required in order to learn new material. Too much of it can cause anxiety and distress, while having too little can be seen in those with ADHD and learning disabilities. At its optimal level, it can be effective in learning, information processing, cognition, perception, REM sleep, and binding senses. To significantly increase it, meditation works as a method to bring the level back up. It is the brainwave of highest frequency.

Beta Waves

Beta waves are commonly observed when we are awake. With the right amount, it can help us focus in school or at work, allowing conscious thought, logical thinking, and is thought to have a stimulating effect. You may liken it to brain waves that are responsible for people to do critical daily tasks, such as writing, reading, Socialisation, and critical thinking. It is also associated with a high level of arousal, in higher beta frequencies. In excess, it can lead to stress, restlessness, and anxiety. Too little of it can lead to frequent daydreaming, depression, or poor cognition. Since it is crucial in our daily tasks, it can help in our problem solving, focus, and memory at its optimum level. To increase it is fairly easy – energy drinks, coffee, and other stimulants can easily do the trick.

Alpha Waves

These waves lie in between our conscious thinking and subconscious mind. When needed, it helps us calm down and relax. Too much can lead to daydreaming, being too relaxed, and the inability to focus. Too little can result in anxiety, inability to sleep, and OCD. To enhance alpha waves, relaxants such as alcohol, marijuana, and depressants can be helpful.

Theta Waves

As long as Theta waves aren't produced in our waking hours, it can be very helpful in our creativity, emotional connection, intuition, and relaxation. Too much theta activity may make people highly suggestible, because of the state of being deeply relaxed and semi-hypnotic. This hyperactivity and inattentiveness are the exact opposite of the effects of low theta waves, which include poor emotional awareness, stress, and anxiety. Depressants help to increase theta waves.

Delta Waves

Delta waves represent the slowest brain waves ever recorded in humans, often found in young children and infants. We tend to produce less of them as we age. Deep levels of relaxation and restorative, healing sleep are associated with delta waves, as with natural healing and immunity. Too much of it is associated with ADHD, inability to think clearly, learning problems, and brain injuries. Having too little can be likened to waking up not feeling refreshed after sleep, and the inability to revitalize or rejuvenate the brain. Depressants and a good night's sleep can help bring it to its optimum level.

Chapter 13 : Types of Memory

Memory is an integral function of the brain wherein certain information is encoded and stored and, under instances where it is needed, retrieved. Memory is an important function in the nervous system; without it, the brain would not be able to remember how to eat, breathe, do certain things, nor to walk or talk. Without having the information stored into our brains as memory, our bodies would be useless and we would be nothing but an empty shell.

In general, there are 3 types of human memory; the sensory memory, the short-term and the long-term memory. These types affect the type of activities one person has regularly. Often, failure or malfunction of a certain memory type would result in memory related ailments, often associated in elderly people such as Alzheimer's disease.

Sensory Memory

The sensory memory refers to the part of the system wherein the sensory receptors store and process the received information long enough for it to be transferred to the nervous system for further processing. It is considered one of the shortest term memory, as it usually takes place for only a short period of time, specifically the travel time of the information from the sensory receptors to the nervous system.

The Sensory Memory, commonly abbreviated as SM, lets the person retain specific information through impressions even after the receptors have been stimulated. SM will allow the travel of the information up to the brain, allowing the sensation to be categorized and associated with the main senses of the body which are touch, sight, smelling, tasting, and hearing. Sensory Memory however, is very short-term as the information is immediately forgotten after a few seconds .

There are sub-types of sensory memory, each classified according to the sensory organ where the information was gathered.

Two of the most studied are the iconic and echoic memory. Iconic memory refers to the memory of the information gathered visually. Studies about iconic memory suggests that the iconic memory of a person is very short so

the brain would barely have the time to process the information received since the storage of such would only last for less than a second.

Echoic memory on the other hand, refers to the memory of the information gathered through the sense of hearing or sounds. Unlike iconic memory, the information gathered through hearing tends to last longer, up to four whole seconds before being completely forgotten. An experiment to test the echoic memory in a person is done by dictating a series of number to them.

If the person is able to retell the series in less than 4 seconds, then there is a big possibility he would get it right and remember everything. If the person waited for a few more seconds however, there is a chance he would have forgotten parts of the series at that point.

Short-Term Memory

The short-term memory, on the other hand, refers to the brain's capacity to store and hold information in miniscule amounts for a short time period. Unlike the sensory memory, short-term memory would last from 20 up to 30 seconds before being forgotten. Often, Short-term memory is associated with the working memory as they have almost the same set of definition and parameters.

Take note that there is a difference between the two however, for while working memory would refer to the active processes you would temporarily store in the brain in order to properly manipulate the information you receive, short-term memory, on the other hand, would refer to temporary storage of information itself.

With short-term memory being temporary information storage, there are limits and duration to this type of memory. For one, only up to about 7 items or information may be stored at a time. There is an idea introduced by the Psychologist, George Miller in 1956, suggesting that short-term memory may be expanded in an approximately seven plus or minus two items at a time.

As a person ages, one would feel symptoms of memory loss. Deficit of this ailment are often attributed in short-term memory, and is visibly observable to people between 55 and 85. Memory ailments which affect the short-term memory include Alzheimer's disease, Advanced Ace, Schizophrenia,

Aphasia, Short-term memory and intelligence, and Post Traumatic Stress Disorder.

Long-Term Memory

LTM or the Long-Term Memory, unlike the SM and STM, would last for such long periods of time. It is also known as the reference memory, and this refers to the information stored and used for daily task performance. Unlike STM where memories would last for about 30 seconds at most, Long-Term memories are very long and could last from days to decades at most.

LTM is classified into 2 types; the explicit and implicit memories. Explicit or Declarative Memories refer to the information your brain looks into for conscious actions and decisions. Such memories could refer to either specific events or any particular knowledge or information that you use daily.

The Procedural or implicit memory, on the other hand, refers to the information your brain uses for unconscious activities, including body movement and the capacity to manipulate objects. Another perfect example of procedural memory is driving or using a gadget such as a computer. In these both cases, learning them at first may have been integral, but they would gradually move on as an instinctive or natural process as you go along, knowing fully well the process of manipulating the car without much conscious effort.

In the following chapters, we will delve deeper into understanding memories. We will find out what we can do to improve our memories, and how our daily lives and activities affect our information storage capacity over time.

Chapter 14: Photographic Memory

Photographic memory refers too much more than just remembering the details of a photograph. It is about remembering virtually everything that a person may be able to recall. This includes not only visual imagery but sound, words, smells, tastes, textures, feelings and even thoughts. Yes, thoughts.

Thus, photographic memory is something truly remarkable in a person. Those who are able to develop it can truly master the art of learning, research, investigation or just paint really good pictures, both literally and metaphorically.

When you develop photographic memory, you become adept at remembering everything around you. You are a master of understanding what is happening within your environment and how you can transform that into useful data.

Now, please bear in mind that having a photographic memory does have certain caveats. For instance, there are folks who confuse photographic memory with having an uncanny ability at remembering facts and trivia. These are the individuals who may remember who won the American League batting title in 1964, but may have a heck of time recalling what they did last weekend.

In such cases, you are witnessing selective memory. We all have selective memory as our brain is able to filter out information based on what we believe to be useful and relevant, which is not getting the heave-ho.

Selective learning plays a pivotal role in learning as we tend to assimilate better that which is interesting to us, rather than what is not. In such cases, you find students who are brilliant at math, but may be flunking language .

On the other hand, there are individuals who are able to recall vast amounts of information with very little detail. They will recall many of the events in their lifetime, but may not do a good job of remembering the actual details of what happened.

For example, you may have an individual who recalls every- thing that happened at a wedding. They know who was there, what they had for dinner, where exactly the wedding venue was, what the weather was like

that day, yet they may very well forget they had too much to drink and caused a scene.

This type of recollection, while selective, only picks out cer- tain parts that the individual believes are worth remembering. Their recollections of the embarrassing moment they caused get stricken from the record, since it isn't a very pleasant memory.

This particular type of selective memory is very common in people who have been through a traumatic experience such as abuse, assault, accidents, and similar events. They tend to delete certain parts of the incident in order to protect their mental state. However, these memories do not get erased; they are merely compartmentalized, meaning they are locked away somewhere for safe keeping.

Compartmentalization is like hiding a bunch of papers that you do not want anyone to find. So, you put them into a pile and lock them away in some filing cabinet. The only problem is if anyone looks through the cabinet deep enough, they will find them.

All of these issues aside, photographic memory is about recalling as much as possible in regard to an event, a block of information, or stimulus. In the case of the drunken guy at the wedding, his photographic memory allowed him to remember what he was doing, even as he was impaired by the amount of alcohol he had consumed.

In the case of a traffic accident, for instance, the victim in the accident may not recall the details of the accident based on the fact that they have a photographic memory. Rather, they may recall the details of the accident simply because they have developed their autobiographical memory.

So, which one is it ?

Well, it can be hard to tell.

Here are signs that you indicate you may have a photographic memory. This list intends to provide you with some general guidelines as to what types of things you need to recall in order to consider yourself a holder of photographic memory.

You Remember Everything About Everything

Alright, this first sign is a little vague. When we say that you remember “everything about everything” we mean that you remember an unusually large amount of information on a given incident, block of information or stimulus.

Consider this situation:

A group of friends goes to watch a football match. Let’s assume that they are all similarly aged, they don’t consume alcohol during the match, and they are all cheering for the same team. Under this scenario, the game unfolds from beginning to end.

Then, every one of the members in the group is interrogated on the details of the match. The friends who have “normal” memory skills would recall the main highlights of the match, such as the goals that were scored, who scored them, and some memorable plays from the match.

Now, the friend with photographic memory would not be able to give you a play-by-play recollection of the match. Such a feat is an unbelievable act that doesn’t seem to be within the realm of human ability.

What you may find is that the holder of the photographic memory will recall very specific details about the match, such as the attendance, the player substitutions, the names of the referees, or even very specific plays that happened over the course of the game.

However, there is a catch. Selective memory may filter out some of the information that a photographic memory will capture. So, they may find that they will recall the goal scorer’s statistics but may completely disregard the commercial broadcast on the jumbo screen .

You are Overly Perceptive

One of the hallmarks of superior memory skills is being overly perceptive. These are the folks that are called “human lie detectors.”

Why?

Well, they say, “the devil’s in the details.” So, these folks are able to pick up on inconsistencies in people’s stories, by poking at those details. They will often ask insightful questions that will challenge an individual to the point where they fall into the trap of their own lie.

This is something that perceptive individuals can achieve because they are able to remember very specific details about the stories people tell. In fact, some of the best liars and con-artists are the ones who have incredible memory skills. The reason for this is because they are able to keep up their lies by not contradicting the details in their stories.

So, if you are overly perceptive such as noticing subtle changes in people's appearance, the objects they have or wear, or perhaps recall very specific details about their lives, then you just may very well have a photographic memory. Just keep in mind that while people appreciate someone who pays attention, paying too much attention may be a bit creepy.

You Often Forget Where You Place Keys, Wallets, Phones, Etc.

This sign may seem rather counterintuitive. After all, how can someone with photographic memory forget anything? But that's the point.

The brain has a natural mechanism that clears out the clutter it deems unneeded. As such, the brain is adept at getting rid of the information that it figures it won't need. In its place, the information that is truly needed is kept. This means that the information that the brain filters as relevant takes a predominant role.

What this implies is that your photographic memory will always be skewed toward the things that you feel are important .

So, if the location of your keys is important, then you can bet your brain will keep tabs on that. Otherwise, you may end up losing your keys, wallet, and even your phone.

In fact, one of the signs of superior intelligence is absent-mindedness. After all, it is almost cliché to see geniuses who are clumsy and forgetful. The underlying concept is that the brain devotes as many resources as it can to the processes that it feels are necessary. However, the conscious mind can also become so focused on one thing or another, that it pulls the brain into a different direction. In a way, it's like trying to have conversations at the same time.

Of course, you can always use your brain power to focus on one thing at a time. This is why multitasking is highly discouraged. Nevertheless, if you are forgetful, you may not just have superior intelligence, but you may also have a photo- graphic memory.

You Recall Faces or Names

Do you commonly recall people's faces but perhaps not their names? Or, do you recall an entire list of names but have a hard time matching them to a face?

This is one of the classic signs of photographic memory. Generally speaking, the brain is able to match a face to a specific event or situation in which it is unable to forget this person. For instance, you may recall the faces of people whom you knew 10 years ago. You may not be able to recall where you know them from or perhaps what their name is, but you don't forget who they are.

Conversely, you may recall a name immediately. You may very well remember the person's entire life story. Yet, if you actually saw them, you may not recognize who they are right off the bat. It is not until they tell you their name that you begin to recall just about everything you can about this individual. In fact, you may end up recalling stories from years ago that others may have completely forgotten about.

Please bear in mind that recalling specific details about events and people is one of the highlights of photographic memory. So, it may be you recall certain events to a point where you can remember the taste of food, the color of clothes and even the feelings and thoughts you had that day.

In short, photographic memory is all about being able to focus on these details down to an incredible level of specificity.

As such, don't worry if you run into people and not be able to recognize them right off.

When You Remember One Thing, Which Reminds of Others

Since the brain doesn't really delete any information, it is stored somewhere in your memory. As a matter of fact, when you "forget" something it is because the brain never actually registered it, or the brain has lost the way to access that information. When you "lose" the path needed to find a piece of stored information, you may end up trying to recall for days about one thing or another. And, it isn't until something else triggers that information, that you remember you had it stored somewhere in your mind.

When this happens, the neurological network in your brain begins to uncover information that it had previously lost in the shuffle. This is when memories about one thing or another come rushing back.

Has this ever happened to you when you are reminiscing about the good old days with your friends and family? Some- one tells a story, and that triggers a flood of memories. In short, one specific comment or recollection “jogs your memory.” Then, you begin to recall these events in such vivid detail as if it had happened just yesterday.

The main focus of this type of exercise is to help your neurological rebuild the access it used to have old files. This is also true when you learn something and then drop it. After some time, a short refresher makes all that knowledge come back.

Isn’t the human mind absolutely brilliant?

Chapter 15: Memory Myths

You were born with a bad memory. Your memory is like a light switch — sometimes on, sometimes off. You have a memory problem beyond repair.

If you agree with any of the above statements, here's some good news: You've been held captive by the memory myths. You can break out of those memory myths that hold you captive and improve the way you remember.

To get on the road to memory improvement, you need to throw away the myths you've heard about memory. Some of these myths are so wrong that simply believing in them adds to your stress level.

These myths include the following:

Believing you will lose your memory no matter what.

Believing your memories are little files of information found in specific parts of your brain.

Believing your memories are true snapshots in time, uncontaminated by all the living you've done.

Believing you don't have to pay attention to remember.

Believing you're too old, too young, or too dumb to improve your memory.

You find out about the most common myths and replace them with the facts about memory. These facts form the foundation for your effort to improve your memory skills .

Thinking You're Losing Your Memory

If you're like most people, you worry about losing your memory. You may forget someone's name in the middle of a conversation or misplace your keys and spend the next half-hour looking for them.

Never fear. You aren't losing your memory, because it's not a thing to be lost. Sweep away this and other myths and discover your memory is a skill that can be sharpened.

Your memories aren't stored in specific corners in your brain. They're widely dispersed throughout your brain After more than 100 years of

research, those experts who study the brain haven't found a specific place where memories are stored.

Many people are fond of using the image of a computer as a way to describe how their brain works. You may be one of these people who says, "My brain is like a computer that's obsolete. I can't access files because my brain's not quick enough, and I have no more room for new information."

Your brain and your memory skills are far more complex than any computer. Your memory skills can improve and aren't pre-programmed like a computer. Memory isn't a hard-wired part of your brain.

Your memory isn't a possession; it's an ability. It's not something you either have or don't have. Think of the word memory as a verb rather than a noun.

You may think, "Well, that's all fine and good, but I was able to remember more in the past." This statement may be true. A person's memory ability can worsen over time in a variety of ways.

Here are some of the reasons why you may experience a decrease in your memory:

- ✓ A poor diet
- ✓ Alcohol or drug use
- ✓ Stress (including anxiety and depression)
- ✓ Medications with memory-depressing side effects
- ✓ Sleep deprivation
- ✓ Difficulty with paying attention
- ✓ Decreased mental activity
- ✓ Utilizing poor memory techniques

Many of these problems can be resolved. You can support your brain's ability to remember by doing the following:

- ✓ Maintaining a balanced diet
- ✓ Taking vitamins and other supplements
- ✓ Avoiding alcohol and drugs

- ✓ Avoiding food supplements, such as MSG and aspartame
- ✓ Exercising on a regular basis, getting some sleep, and relaxing.

Memory Is Like a Filing Cabinet

If your memory is an ability and not a thing, what kind of ability is it? Is it like riding a bicycle, driving a car, or cooking tonight's dinner? Sort of. However, memory is far more complex and underlies each one of these activities.

Memory is the basis for everything you do and everything you learn. You can't learn without it. Some people who have suffered severe brain injury have also lost the ability to remember in a variety of ways, depending on what area of the brain they injured. Memory skills work together as one memory system. If part of that system is damaged, they can still remember, but with specific limitations. For example, they may not be able to remember the shapes of objects as well as they did before the brain injury. Yet, they retain the ability to remember in ways the healthier parts of their brains allow .

You have basic memory abilities, such as remembering what you hear, what you feel, and how you move. You also have complex types of memories. You remember shapes of various objects and faces, and you remember tunes, words, and numbers. Finally, you have even more complex types of memories, such as whole sentences, whole pieces of music, mathematical equations, the names of the capitals of the countries of South America, and so on.

Even the simplest of tasks involves multiple memory systems. When you drive your car, you're relying on a variety of different memory abilities. As you pull up to an intersection, you remember that it's a time for caution. You see a red, eight-sided sign and remember it's a signal to stop (visual/spatial memory). You notice the red sign has the word Stop painted on it, and you remember the word means you must make sure your car comes to a complete stop (advanced verbal memory). You automatically remember you need to remove your right foot from the gas pedal and place it on the brake pedal to bring the car to a halt without locking your tires (movement, touch, and spatial memory). Then, after you've come to a complete stop, you abide by the rules of traffic, remembering the driver to the right has the

right of way if you arrive at the same time (conceptual memory, involving many systems).

You acknowledge it's his turn to go ahead, so you look at him and wave (complex conceptual memory). He drives off. When it's safe, you take your foot off the brake and place it on the gas pedal, gradually applying pressure (movement and touch memory).

Even in this simple experience of coming to a stop at an intersection, you're accessing multiple systems of memory. Almost everything you do involves systems of memory rather than one discrete memory.

Memories Are Snapshots, and Cameras Don't Lie

Who you are in the present determines what you remember about your past. Think about that, and you can see why you're sometimes surprised by your memories.

Have you ever gone to visit the elementary school you attended as a child and found yourself shocked it's a lot smaller than you remember? Have you ever reflected on an early romance gone awry and found yourself remembering many of the idealized high points and few of the low points?

Just as memories aren't things, they also aren't etched in stone. Your memories aren't snapshots in time, to be remembered with complete accuracy later in life. Your memories are reworked, or re-remembered, from your perspective in the present. You look back at your life through the filter of all the experiences you've had since the events you're trying to remember.

When you think about your return to the elementary school, your memory is affected by who you were then and who you are now. When you were a child, your school was "big" to you.

In this sense, your memory that your school was big represented who you were at the time. You were smaller then, so everything looked larger to you. But you've probably even exaggerated the memory by making the school larger than it actually was because you experienced so much at that school. It was one of the primary centers of your childhood experiences.

Remembering becomes even more complicated when you try to remember events surrounding a romantic relationship. You look back at the love affair,

remembering all those passionate moments. Somehow, you overlook the events that led to your breakup. You don't remember the whole picture because your memory of those passionate moments is so vivid.

Not only were those romantic experiences very important to you, but you remember them from the vantage point of the present. You probably exaggerate those memories of the high points if you're between relationships or in a bad one now. On the other hand, if you're in a good relationship, your memories of a past romance may not be as exaggerated.

Now that you understand memories aren't simple snapshots of information, consider the highly controversial subject of repressed memories. Repressed memories are traumatic memories that have been pushed back into a person's unconscious. Over the past 20 years, some mental health professionals have tried to help their clients discover memories that may have been repressed. A therapist may try to help the person fill in the "gaps" in their memory of childhood .

You're Too Old, Too Young, or Too Dumb to Improve Your Memory

You likely have heard someone excuse himself or herself for a memory lapse by saying, "Sorry, I guess I'm having a senior moment." However, being a senior doesn't mean you're doomed to lose your memory.

Yes, psychologists have shown people do experience some decline in memory as they age. But this decline isn't as great as is popularly believed. Neither do all seniors experience memory decline at the same rate.

Though some evidence suggests that visual and spatial memory erodes faster than verbal memory, everyone doesn't lose this memory at the same pace.

Speaking of the younger set, whether you have a teenager at home or are a teenager yourself, you may have heard people say, "He just doesn't seem to have his head screwed on tight." Being a teenager, or being a child for that matter, doesn't mean you have to expect memory lapses.

Children and teenagers can improve their memory skills not only through maturation, but also through application of memory techniques. Why not get a headstart and do better in school?

Chapter 16: Attention is the Pathway

A positive effect of having a clear and concise memory is by being able to fully concentrate on a particular task. Eliminating distractions in everyday life is truly a challenge for anybody, and for the most part, people find themselves in trouble, figuring out a way to concentrate and focus on a task. Here are some things that you may try doing in order to sustain your focus and retain your concentration.

Eliminate distractions

People tend to have terrible digression and short attention span to whatever task they are doing when there is something else that keeps their attention away from the task at hand. It could be through a text or call, a television program, a particular game---anything at all. The first and probably best thing you can do about this is to remove all the potential distractions. Shut off your cellular phone and tuck it far away if you have to, or perhaps turn off the TV. This way, you would have your undivided attention to what you are doing and not find your mind floating somewhere else.

Take a break once in a while

There may be tons of things for you to do and that could be overwhelming. There are even times when somebody is so overwhelmed by the swamp of things he must accomplish, his brain would tend to retaliate and cease to properly “function,” causing you to not do nor accomplish anything in the end. Studies show that a person’s attention span to a particular activity lasts for 2 hours at most.

It is therefore essential for you to take a short break---probably going away from your task laptops or books to get coffee somewhere. This way, you would not feel stressed and would have enough time to process and sort out your thoughts as you get back to the job to be done.

Create lists and reminders

One of the best ways to ensure you will remember everything is by creating a to do or reminders list. Doing so will allow you to see all the things you need to accomplish. Place post it notes on the important parts and cross out the things you have accomplished. Doing so will allow you to have a clear

overview of the things you were able to accomplish and the rest of the items that still need to be done.

Time Management

Proper time management is always a part of maintaining concentration. Setting a particular timeline to accomplish a particular task encourages a person's memory and skills to do their best in finishing an activity, warranting their full attention in order to do so. Start by accomplishing tasks that would consume the shortest time; this way, you would be able to accomplish a lot in a short amount of time, and be able to concentrate and give your best in longer tasks.

Proper diet

Believe it or not, a balanced diet plays a very important role in memory enhancement. The nutrients you would get from food stimulate your memory. A person with insufficient amount of nutrients in the body would result to him being sluggish, and therefore unable to concentrate on what he has to do. Further, the idea of being hungry and the desire to eat is often a very strong distraction that will fully disrupt your concentration. Therefore eating (this should be healthy and in moderation!) should take precedence once you feel the need to do so. After all, there goes a saying that "an empty stomach leaves an empty brain," and you wouldn't want to have that.

Have enough sleep daily

This is similar to that of the food distraction. Lack of sleep causes low energy, therefore letting the body feel very sleepy. Your brain tends to float to outer space without as much energy that it could have had. Never be worried that you feel sleepy in the middle of a task! Take a power nap---usually about 10 minutes to an hour, so you can regain your energy to work and concentrate some more.

There are plenty of other things that you can do to guarantee concentration. Included in the list are balanced and appropriate exercise, self-challenge and medication. Just make sure that all these, especially the one with medication, to be consented and recommended by a medical professional for you to experience optimal results.

Chapter 17: Harnessing the Power of Your Brain

Your brain is an amazing storage box that has the surprising ability to adapt and change even into old age. This amazing ability is known as neuroplasticity. Your brain has the ability to form new pathways, modify open connections, and get used to ever changing ways with the help of the right stimulation. When it comes to learning and memory, your brain reshapes itself and holds true with its amazing ability. To increase your cognitive ability, boost your ability to learn new facts and figures, and improve your memory, you can harness the natural power of neuroplasticity.

Do Not Skip on Exercise or Sleep

Your brain and memory relies on a nutrition-packed diet and on sleep to perform at its best. When you care for your brain with healthy habits and a good diet, then the ability to remember things increases with it. Treat your body with good health and increase your ability to work and recall information. Make a habit to do physical exercise because that increases oxygen to your brain and decreases the risk for disorders that causes memory loss, such as diabetes, etc. Exercise increases the effects of brain chemicals that are helpful and protects the brain cells.

Healthy exercise, diet, and sleep are also necessary for improving memory. If you have lack of sleep then your brain finds difficulty to operate with its complete capacity. Due to lack of sleep, your creativity, problem-solving abilities, and critical-thinking skills become affected. Sleep deprivation is a recipe for disaster for your studies and work. Sleep is necessary and is a memory-enhancing activity and a basic way to improve your learning .

Make Time for Friends and Fun

The best way to improve your memory and relax your brain is to make friends and have fun with them. A life full of friends and fun has cognitive benefits. Our healthy relationships are the ultimate memory boosters that stimulate our brains. When you interact with your friends and family it is the best exercise for your brain.

The next tip for improving memory is to take a medicine of laughter as much as you can as it will keep your brain healthy and fresh. Listen to jokes

and work out punch lines that activate the learning and creativity areas of your brain.

Keep Stress in Check

Stay away from your brain's worst enemy and that is stress. Stress destroys brain cells and damages the power to think and remember anything. Improve your focus, creativity, memory, concentration, and learning skills by doing meditation as it decreases your stress. Meditation leads you to feelings of joy and equanimity and boosts your learning abilities.

Eat a Brain-boosting Diet

As your body needs different healthy fuels to keep it strong and powerful, so does the brain. For improving memory, eat a healthy diet based on vegetables, fruits, whole grains, and fats that boost your memory such as nuts, fish, olive oil, and almonds. Omega-3 fatty acids are important for your brain's health.

Give Your Brain a Workout

Memory is like a muscular strength, it requires you to use it. For better processing and to remember information better, you should work out your brain more. These brain exercises break your routine and challenge you to use it and develop new brain pathways. The activities for your brain workout are: playing musical instruments, enjoying games, knitting, tennis, and all those activities that challenge your brain also includes spatial-temporal reasoning and creativity .

Improving Memory

You can employ a number of techniques to improve your memory. But because you're a complex human being, no one method alone can do the job. You need to do a number of things all at once to improve your memory.

Never fear! Improving your memory isn't difficult if you do the following:

- ✓ Understand the facts and myths of how your memory
- ✓ Make sure that your brain is capable of remembering
- ✓ Avoid memory depressors
- ✓ Keep your mind sharp by challenging yourself intel- lectually.

- ✓ Learn memory techniques, such as mnemonics
- ✓ Apply your memory skills on the spot, such as at school, at work, or in social situations

Discovering what your memory skills are

To improve your memory, you need to know what memory is and isn't. You need to understand the facts about your memory and discard the myths.

Some of the following facts may surprise you:

Your memories are constantly reshaping throughout your life

They aren't snapshots of information, frozen in time. Throughout your life, you go through all sorts of changes: You mature, learn from previous mistakes, and even get tainted by unfortunate experiences. Your perspective changes over time, and the lens that you use to look back at your memories colors those memories.

Your brain is far too complex

It is always changing, which affects the way you look at your memories .

They're subject to modification

Everything you experience affects your memories constantly

Your memory is a skill that you can cultivate and improve upon, no matter what your age, intelligence level, or socioeconomic class

Your memory isn't something that you have or don't have. It can't be lost or found like a memory chip. However, if you do incur damage to your brain, the amount and type of damage can limit your memory skills .

Your entire brain works together as one system

Each memory has input from a wide variety of different parts of your brain. Memories aren't stored in one place in your brain. Your brain codes memories like a system of different units. Some areas of your brain have unique talents in coding memories that correspond to shapes.

Your brain possesses several key parts that play a major role in forming the foundation of your memory.

One the key parts of your brain is called the hippocampus, which moves short-term memories into long-term storage.

The short-term memories that you move into long-term memory depend largely on three factors.

Your attention. Generally, a measure of short-term memory.

The meaningfulness of the information. The more meaningful or relevant the information, the greater your chances of remembering it.

How well the memory fits into what you already know. A foundation of memory helps with new learning.

Your memory depends on your attention

Your memory simply doesn't work unless you're paying attention. But when you're paying attention, your brain is capable of extraordinary mental gymnastics. If you protect and care for it, you can possess keen memory skills .

Feeding your brain properly

Your ability to remember depends on a number of factors, first of which is maintaining a healthy brain through adequate nourishment. To ensure your brain works at an optimum level, follow these guidelines:

Maintain a balanced diet

What you eat affects your brain's chemistry. Eating the right foods at the right time gives your body the building blocks to manufacture brain chemicals called neurotransmitters. Neurotransmitters not only affect your mood and ability to think clearly, but also your ability to remember.

Take the right supplements

Vitamins and herbs can support your brain's ability to produce good memory skills. Specific vitamins, such as some of the B vitamins, help form the building blocks for healthy brain chemistry. Other vitamins, such as vitamin E, help your brain cleanse itself of bad chemicals. It helps combat the destructive but subtle brain-wasting changes, such as free-radical damage, glycation, and homocystine.

Herbs, such as gingko, have been used to enhance the circulation of blood in the brain Gingko dilates the capillaries (small blood vessels),

which can help your brain cells get the vital nourishment they need.

Get regular exercise

Exercise enhances your memory because it helps your brain receive the nutrients that it needs. Every time you exercise, you increase your respiratory rate, your metabolism, and your energy level.

Avoiding food, drink, and drugs that depress memory

Foods, drinks, and chemicals that your brain is exposed to have major effects on your memory. I explore this in detail in.

If you eat junk food loaded with sugar, you set yourself up to crash, finding yourself full of anxiety and short of short-term memory. Similarly, if you drink too much caffeine, the liquid anxiety scatters your thoughts and shatters your memory ability. If you consume caffeine and sugar on an empty stomach, your mood and memory skills will plummet quickly.

Keep these two guidelines in mind:

1. Avoid heavy amounts of sugary foods (like candy, cookies, and so on).
2. Minimize caffeine (and never drink coffee on an empty stomach).

If you treat your brain badly, your long-term memory can erode, and your brain will reward you with few memory skills.

You can avoid devastating your brain and memory by staying away from the following substances:

Alcohol - It kills not only your brain cells but also what they produce — a good memory. Alcohol causes your thinking skills, which are so critical to a good memory, to erode away.

Drugs - Marijuana and other drugs can cloud your memory so badly that you lose the ability to focus. The chemical found in marijuana, called tetrahydro- cannabinol (THC), can really put a damper on your short-term memory as well as your motivation.

Neurotoxins - These poisons include petrochemicals, solvents, and herbicides. They kill brain cells when you’re exposed to them.

Food “enhancers” - Monosodium glutamate (MSG) and artificial sweeteners are damaging to brain cell operations

Chapter 18: Brain Training & Nutrition

This chapter gives you a complete guide on how you can train your brain and improve your memory. In the field of brain fitness and improvement of memory, the topic of brain training is the most popular and amazing topic. To increase our memory, we are seeing new tools and tricks.

The scientific brain fitness games or brain training software or online puzzles are the powerful tools for your brain training. Don't be scared with these options. If you can't buy expensive brain software, we do have an alternative to guide you on how you can improve your memory.

Start practicing brain training exactly at your own pace in your spare time. The brain training techniques are easy, fun, and amazing.

Brain Training

Keep Your Senses Working Through the Day

The first brain training tool to improve memory is when you see a mesmerizing picture, make a note of it in your brain about its colors and the things you like in the picture. Take a moment to look at the picture and try to memorize every detail about it. Recall the image throughout the day. The more you think about every detail of the picture, the easier it is for your brain to keep the memory and remember it at a later time.

Focus on One Thing at a Time

In our everyday life we try many things to remember, but due to distractions we fail to focus on one thing. In the second brain training tool, it teaches you how you can memorize things for a long time .

To remember some facts and figures in your life then turn out all other distraction and keep your focus on the one fact. This kind of thing can be a chat you had with your family or friends, or employee, or a magazine article you read. By focusing on one thing, you are giving a boost to your memory.

Practice Deep-Breathing Exercises

Exercise is a powerful tool that keeps your brain active and strong. Train your brain by practicing deep-breathing exercises. It will truly relax your senses and your brain will start to work easily. It's an amazing way to recall

a situation when you start trying to remember things too hard and your body tends to tense up and your mind's stuck. Try to practice deep breathing daily as it will relax your state of mind and it will be easier for you to memorize things.

Play Word Association Games

The next tool is a fun tool for brain training. Associate things with something you want to remember later and you don't want to forget. This technique is very useful, especially when you have difficulty in remembering people's names.

Do a Crossword Puzzle Everyday

A more fun and amazing training technique, if you are aware, is crossword puzzles. It can be fun for you, but if you have never done crossword puzzles then you will be amazed at how much fun they are and how they will encourage your brain. It is easy to purchase in dissimilar ability levels and provides you and your brain with amazing and challenging levels.

Brain Nutrition

You may not see the direct link between diet and brain health since what we put in our mouths goes downstream and not upstream. But what we eat has a lot to do with how healthy our brain is. Granted, food does not travel up to the brain and its nerve cells, but blood carries nutrients and oxygen, and if that blood is tainted, then how can it be an effective carrier of good vitamins and minerals ?

The link between diet and brain health can be summarized in this continuum:

Bad food = unhealthy and sluggish bodies = disorders + diseases = slower brains

"What we ingest is also fundamental to how we think and feel," say doctors.

"The brain is an incredibly active furnace, consuming 25% of the glucose and oxygen we take in. It burns glucose as its sole fuel, and yet it has no storage site for it…The brain also needs a range of other nutrients…Even minor nutritional deficiencies can be associated with changes in mood. Memory loss, confusion, depression and other mental disorders in the

elderly, once attributed simply to aging, can also be consequences of a poor diet."

The next question logically would be, so what foods can we eat to make our brains healthier including sharpening our memories and "making learning a breeze?" We'll focus on the basic foods groups – fats, carbohydrates, micronutrients and other kinds of foods that would supply us with the vitamins and minerals we need.

The general dietary guidelines to optimize brain power.

Fats

You may have heard someone say once upon a time that nuts will improve your memory. There's some truth to that statement. Our bodies need essential fatty acids (EFA). Seeds, raw or dry roasted nuts are a source of EFA.

Your eating routine must include walnuts, pumpkin seeds, flax seeds, and other types of nuts. If you can get cold pressed oils from these nuts, this would be an added benefit.

Omega-3 essential fatty acids can be sourced from salmon, sardines, trout, tuna, herring and mackerel. Many people do not like fish, but changing our eating habits to include fish at least once or twice a week to start makes for a healthier brain. Wild fish, rather than farmed, are better. Needless to say trans-fats must be avoided .

You can also get EFA from avocados, fresh coconut and extra virgin olive oil. Meat is also an excellent source of EFA, but as much as possible try to get meat and dairy products from animals that were raised naturally. People in the industry call it "free range" animals; this means that the animals were allowed to run free in open fields and grasses so their diet (and what we eventually consume), are rich in EFA.

Carbohydrates

We need carbohydrates to replenish our supply of brain fuel glucose; this is why we should eat frequent small meals instead of three square meals a day. The inability to concentrate and feelings of weakness can result from hunger or you allowed too much time to lapse since your last meal.

Corn, potatoes, winter squash and cooked or juiced carrots and beets are nutritious and high-glycemic foods; so are whole grains, cereals and crackers.

The only problem with consuming foods rich in carbohydrates is we could raise our sugar levels unexpectedly, but this can be corrected if we combine them with sources of protein. Some examples include egg with toast, soy burger with corn and salmon with potato.

Doctors offers this “carb” wisdom: Foods vary greatly in carbohydrate content. For example, a half cup of carrots has only 3 grams of carbohydrates; a cup of cooked macaroni, 52 grams; 2 cups of popcorn, 12 grams; a plum, 7 grams.

Micronutrients

The general idea is that anything that has leaves and is dark colored is a good food with sufficient vitamins and minerals. Fruits and vegetables have high levels of micronutrients. They are rich in antioxidants that help our bodies fight free radicals, culprits of most of today’s diseases and disorders.

Nutritionists recommend five daily servings of fruits and vegetables but cancer specialists recommend increasing that portion to nine to ten servings. Use a small fruit or half a cup of cubed fruit as your serving guide. Go for vegetables that are in season and are grown in organic farms .

The idea of nutrition playing a vital role in brain health is clearer when the brain exhibits signs of a mental disorder. Depression is a good example to cite. While there are accepted medication and treatment protocols for this disease, the role of food and nutrition has to come into focus as well. This is particularly important since it is known that people who are depressed generally have little interest in eating or have erratic appetites.

When a person is suffering from depression, nutritionists believe that to have a brain-healthy diet, all kinds of artificial and fast foods should be immediately eliminated: alcohol, simple carbohydrates, white flour products, artificial sweeteners and caffeine. Eliminating these substances will improve the chemical balance of the brain. The major culprits are fried hamburgers and French fries since they block arteries and small blood vessels, and thus interfere with blood flow.

Individuals who have nutrition deficient diets take supplements to make up for the gap that results when not enough fruits, vegetables and grains are consumed. Before going on a supplement program, you must speak to your physician and nutritionist so a thorough assessment can be made of your health.

Chapter 19: Improving Your Study Skills

This chapter covers all the study skills that improve memory. Every student desires the best study skills to achieve their goal. The best study skills are:

Reduce Interference

Forgetting something is known as interference. When the new material is similar with the previous learned information then the interference that occurs comes from your study skills. Follow the given strategies to enhance your study skills:

Make it Meaningful

Make the information more meaningful by making sense of the material you need to study; it will reduce your interference and helps you remember things. The more you think around a subject, the less demanding it is to learn new material identified with it. Case in point, experienced chess players have the capacity to retain new moves more rapidly than other individuals who are not acquainted with chess.

To my mindset, the guideline of nature is one of the numerous profits of being a long-lasting learner inspired by an extensive variety of subjects. Something you learn today may help you learn related material eventually. The second way to remember information is to convert it into rhymes.

The rhymes will make the information more meaningful and easier to remember. Patterns are another way to make information more meaningful the same way rhymes do. For example, identify repeating patterns or rules that can help you quickly memorize subsections of a list when learning a list of codes .

Avoid Studying Similar Subjects Together

When you study similar subjects together then interferences increase. For example, if you want to study the disorders of eyes and the disorders of ears together, then the interference increases because their names are somewhat similar.

Space It Out

The best amazing study skill is to space out the studying for any particular subject. Most students try to learn all the study material in one study session in what is called cramming. In the same way that separating the material of a few separate subjects into different, separate study sessions improves memory for the material, separating the investigation of a solitary subject does likewise.

There are at least three reasons why this spaced learning is better than cramming:

The Limits of Attention

There is a limit for how long you can focus on something. Don’t try to learn too much at once, keep your ability to concentrate on the facts and figures as you are learning.

Consolidation During Breaks

When you take breaks during studying, your brain strengthens. The less consolidation of memories occurs when you try to learn all the material at once and take no breaks during study.

Differing Contexts/Mood

Environmental context affects memory. When you want to focus and concentrate on your subject then take different contexts for study. Take breaks during a study session. It will boost your memory.

Use Whole and Part Learning

This is the best study skill that teaches you when to take a break during a study session. With this approach, you first use the whole method to get a good grasp of the material. In other words, read straight through everything carefully once or twice.

Then you separate out the more difficult sections for extra study and reinforcement.

First go straight through all the material a couple of times quickly. Then break up the material into logical parts that you study separately. At the end, go back and review everything from beginning to end. With this methodology, you separate the material into areas and start contemplating the first part.

At that point you proceed onward to mulling over the second part, yet you ponder the first part again alongside the second part. At the point when considering the third part, you likewise think about the first and second parts.

You proceed along these lines all through the material. This dynamic taking in methodology keeps you from overlooking the material you learned and it additionally helps you sort out all the material reasonably in your mind.

Progressive Part Method

With this approach, you break up the material into sections and begin studying the first part. Then you move on to studying the second part but you study the first part again along with the second part. When studying the third part, you also review the first and second parts. You continue this way all through the material. This progressive learning approach prevents you from forgetting the material you learned at the beginning as you go along and it also helps you organize all the material conceptually in your brain.

Recite It

Recitation is another good skill for study. It means recalling the information you just learned. When studying your material, take a pause and look away from the material and try to remember the facts that you learned before.

If you feel you forgot it, then go back and read it again. To recall the material for studying, the easiest way to memorize a chapter is by giving headings to each chapter. For reciting the material you just read, get with a partner and recite it together. Recitation is the best study skill to remember things for a long period of time.

Use a Study System

The study system is the fifth best skill to improve memory and for study skills. To approach the study material is a standard method. SQ3R is the best study system. It is an acronym that is based on the steps for the system of Survey, Question, Read, Recite, and Review.

Survey

In surveying, you just need to understand the framework of the information. Read the important structural parts of your study material quickly without going too deep.

Question

When you complete your survey, just go through the parts you just learned and make questions in your mind that you ask yourself.

Read

Peruse straight through everything without taking notes. Make a point to read in detail through the diagrams, section synopses, and so on again also. Utilize velocity perusing strategies, particularly amidst your first go through the material, to spare time and expand cognizance.

Recite

With the information you just read, ask questions and answer yourself. Read all the chapters and read each heading and subheading.

Review

The survey step ought to take just a couple of minutes. The system comprises of discussing some way or another through the material once more. The more times you can survey before your exam, dispersed out over hours or days if conceivable, the better you will probably recall the material.

Chapter 20: Neuro-Linguistic Programming Tips

Neuro-Linguistic Programming, or NLP, provides practical ways in which you can change the way that you think, view past events, and approach your life. It is the secret behind the success of many people. It is an art to achieve goals without any negative implications.

NLP combines all the means of communication that includes verbal language, body language, and written language. By using NLP techniques, your memory can be increased. Recalling is constrained in light of the fact the human personality is constantly turned on and we all experience talking to ourselves among others more often than not. This is one of the foundations for memory misfortune, as one is not ready to review a certain exchange in light of the fact that at the time he was additionally having an extraordinary talk with himself in his brain.

Gain from the past—history is a ceaseless cycle; antiquarianism helps us comprehend the past with a specific end goal to anticipate and comprehend the future better. In one's home life, the same may apply, particularly since people are inclined to rehash their mix-ups. To pick up control over your life and propel forward, you must kick back and examine why you may have fizzled in a certain assignment. Ask yourself how you can manage the undertaking and what prompted the past disappointment. Along these lines, one can enhance his learning capacities and coordinate better into society.

People are sensorial animals and they utilize all their faculties as a part of a request to remember and review a bit of data. NLP advances by gaining experiences by captivating all the faculties. First, one can utilize visualization. Diagrams, outlines, pictures, and even words themselves are simpler to review than theoretical ideas. Utilize this basic NLP strategy as regularly as you can.

Get to Your Visual Prompts

Access the outwardly essential piece of your cerebrum by gazing upward and to one side. You see when you turn upward and to one side you get to your visual memory. These are the pictures and films that have flashed in your brain from the past. You can do this deliberately at this time to

discover things you lost. An Alternate Approach to Utilize This Methodology is by being a Part of a Discussion.

Whenever you talk with somebody, make sure to play with the visual method. When somebody is identifying with you, begin to make pictures in your brain. Listen to individuals as though they were letting you know a story. Furthermore, as they recount to you their story, manufacture a movie in your brain. Add visual pictures to the story. The wealthier the pictures, the simpler it will be to recollect.

Work on Changing Your Physiology

Make a point to address somebody by keeping your head up and shoulders back. Remain straight when conceivable. This makes it such less demanding to gaze upward and make pictures. Also since your memory needs to go through the picture-production enchantment that originates from turning upward and to one side, it is the place you need to be. Work on seeing the world in an unexpected way. Utilize the muscle between your ears. You've got a standout amongst the most valuable blessings the world has ever known: your wonderful personality.

Chapter 21: Balancing Your Mind for Memories

Have you ever shaken a snow globe and watched the snow swirl around the town scene? You can hardly see anything until the snow settles. Imagine your mind and anxiety in the same way. Recalling memories is next to impossible until all the anxiety settles.

You will discover how to defuse stress, calm your mind, and get enough sleep. With a balanced mind, you'll be able to build the memory skills you expect of yourself.

Defusing Stress

You have to be prepared to remember. Your mind must be receptive to new information for it to have an impression on you.

If you're unbalanced because of stress, depression, or lack of sleep, your mind is less receptive to form- ing new memories.

Your anxiety can be a double-edged sword. When you have too much angst, you'll probably have a hard time remembering anything useful for an exam or presentation. Your test anxiety may be so overwhelming that you create a self-fulfilling prophecy and allow your worst nightmare to happen, namely, that you'll do poorly on the exam.

On the other extreme, if you go into the exam with a lazy attitude, seemingly not caring if you pass the exam, you may not pass it. If you aren't the least bit anxious, you probably can't stay alert enough to remember information

you need for the exam. In this case, your anxiety is analogous to fuel, and you're running on empty. Anxiety isn't bad in itself. You just don't want too much or too little of it. You need a balance. You need to be focused and motivated, but not overloaded with stress.

If you feel a great deal of stress, you're probably making absent-minded mistakes, such as forgetting your keys or turning down the wrong street on the way to work.

Negative emotions and preoccupations with personal problems can very easily distract you, making recalling memories extremely difficult. If you're suffering from severe stress stemming a major life event, such as divorce or the loss of a loved one or even the loss of your job, you can be so overwhelmed that paying attention to your surroundings is the farthest thing from your mind.

After you deal with the grief and sadness of your loss, your memory improves because you're freed up from the distractions of your loss.

Relaxing to Charge Up Your Memory

You can induce the relaxation response by a variety of techniques. Most relaxation techniques that you can use involve letting go of sympathetic nervous system arousal by shifting your attention away from the anxiety provoking aspects of your situation.

By bringing the finger up for attention, it's hard to get it out of their mind. But after we go on to another subject, I come back to the subject and ask, "Well, my guess is during the last ten minutes, you haven't been thinking of your right index finger."

They say, "Well no, I haven't. We've been talking about other things."

I say, "That's just the point. As you refocus your attention, instead of trying hard not to think of that difficult event, you'll be able to move on."

Because trying "too hard" to relax actually makes you more stressed, letting go allows your body's ability (through the parasympathetic nervous system) to calm down. Letting go can be counterintuitive. If you try to relax, you'll probably stay tense. Letting go is a passive process. By allowing yourself to relax instead of trying to relax, you take the pressure off yourself .

For example, try not to think of pink elephants. By forcing yourself to keep pink elephants out of your mind, you fight with yourself and invariably think of pink elephants. Now focus on something else, such as blue kangaroos. Tell yourself if pink elephants pop into your mind, so be it.

You probably won't think about pink elephants now, because your attention has shifted and you let go of the pressure that you put on yourself to keep pink elephants out of your mind. Similarly, if you try not to be tense, you'll probably make yourself tenser. By focusing on relaxation, your tension can

melt away because you have shifted your attention away from what was making you anxious.

To achieve relaxation, you can employ any of the following techniques. These techniques involve shifting attention and letting go, and they're generally referred to as relaxation techniques. Some examples include progressive relaxation, self-hypnosis, visual imagery and meditation and prayer.

All these techniques involve a focus on breathing. Your breathing does more than just keep you alive. Breathing right is the gateway to the relaxation response.

Don't take the importance of breathing for granted. If you breathe at different rates, you'll experience different emotional states. Your breathing rate naturally speeds up when you're experiencing stress. The muscles in your abdomen tighten up, and your chest cavity is constricted.

Because you have a cardiovascular system, meaning a system in which your respiratory system and circulatory system are connected, rapid breathing makes your heart rate speed up and thus makes you more anxious. On the other hand, if you slow down your breathing, your heartrate generally becomes more relaxed.

Most people breathe from 9 to 16 breaths per minute when they're at rest. Panic attacks often involve as many as 27 breaths per minute. Accelerating breathing alone can cause anyone to experience many of the symptoms associated with a panic attack, including numbness, tingling, dry mouth, light-headedness, and short-term memory problems .

Slow down your breathing rate by breathing "abdominally," and notice that your belly rises with your inhale and drops with your exhale. This is because as you breathe abdominally, the large dome-shaped muscle that divides your thoracic cavity from the abdominal cavity below expands and contracts.

When you exhale, your diaphragm moves upward and your abdominal muscles contract, helping get rid of stale air or carbon dioxide. When you inhale, your diaphragm contracts, pulling it down, and your abdominal muscles relax. This al- lows for the expansion of your lower lungs.

Your abdominal/diaphragmatic breathing allows you to get more oxygen in and more carbon dioxide out. Because the en-tire process of breathing is to oxygenate your blood and expel waste air (carbon dioxide), diaphragmatic breathing allows you to get more fuel (oxygen) to each organ, especially your brain. Try abdominal breathing the next time you feel anxious. Notice how your anxiety drifts away and your short-term memory returns.

Pulling Out of the Dumps

If you're down in the dumps and depressed, your memory skills become depressed. When you're preoccupied with your own pain, you're less present to form new memories or recall memories formed before you were depressed.

The reason for ruling out depression is that if you're de- pressed, your memory scores will be depressed. Psychologists like me have to be sure that the person has a true memory problem and not a hidden symptom of depression. Because depression itself causes problems with memory, treating depression usually increases a person's memory ability, but how much isn't clear until depression is alleviated.

Think of depression as a drag on all your cognitive abilities. If you're depressed, you're less able to concentrate and pay attention to important information, and you're less motivated, less energized, and more prone to absent-mindedness.

What's more, you run the risk of distorting your memories by putting a negative spin on all of your experiences. Your colleague who isn't depressed may regard what you remember as an unfortunate incident at work as nothing of importance. But for you, the supposed unfortunate incident, much of it fabricated by the negative spin, sticks with you as if you've stepped on gum.

Getting a Move On

This may sound far-fetched, but exercise is good for your memory. It helps your brain receive the nutrients that it needs. Every time you exercise, you increase your respiratory rate, your metabolism, and your energy level. Exercise not only calms you, but it also helps you be more alert.

Your brain, in comparison to the rest of your body, is a high-end consumer of glucose and oxygen.

By exercising, you increase your blood flow, which contains the glucose and oxygen you need to stay healthy and focused.

Exercise maintains healthy organ systems that support your brain, such as your heart, arteries, and lungs. These systems need to be in good shape to support the constant demands of your gluttonous brain. Your cardiovascular system is called that for a reason, meaning your heart and blood vessels working together. As your breathing capacity increases, so does your heart capacity. They're connected. Better circulation means better blood flow to your brain. If you exercise regularly, you feel more invigorated and alert.

This isn't just because of the more efficient blood flow; it's also because of the rich nutrients your blood brings to your brain. Exercise changes your brain's chemistry that's so critical to how it communicates within itself. It stimulates the release and absorption of several neurotransmitters.

You may be familiar with a "runner's high," an endorphin rush resulting from your brain's ability to medicate and reward itself. Endorphins are neurotransmitters, which are like opium without all the addictive and destructive consequences. They enable you to feel calmer and better able to concentrate for hours after you exercise. Better concentration means better ability to remember.

Aerobic exercise not only has a calming and antidepressant effect, but also helps your sleep. To get the best sedative effect, exercise 3 to 6 hours prior to sleep. This time gap allows your body temperature to drop well before you go to sleep .

Getting Some Sleep So You Can Remember

Too little sleep can set you up for a pack of memory problems, faulty short-term memory, lack of clarity of mind, and gen-eral slowing of your ability to think.

If you practice all the following methods, you can improve your sleep:

Don't do anything in your bed other than sleep (except for sex)

Don't eat, watch television, balance your checkbook, discuss finances with your spouse, or argue in bed. Make your bed carry only one association-sleep.

Don't try too hard to go to sleep

You end up frustrating yourself and having more difficulty sleeping. If you can’t sleep and you find yourself tossing and turn- ing, get up and go to another room.

Don’t drink large quantities of liquid at night

This lowers the sleep threshold and causes you to wake up to go to the bathroom.

Don’t work on the computer into the late evening or use any other source of bright light.

Do all planning for the next day before you get into bed

If you think of something you need to remember, get up and write it down. Tell yourself that you’ll post-pone thinking or worrying about anything until the next day.

Avoid all daytime naps

Think of naps as a way to steal sleep from the nighttime.

Try eating a light snack with complex carbohydrates before bed

Eat foods rich with tryptophan. Don’t eat anything with sugar or salt before bed. (I offer eating suggestions later in this section.)

Avoid protein snacks at night

Protein blocks the synthesis of serotonin and, as a result, promotes alertness.

Exercise 3 to 6 hours before you go to bed.

Use earplugs if noise bothers you.

Don’t drink alcohol before going to bed.

Try relaxation exercises to help you fall asleep and/or go back to sleep if you wake-up during the night

See the section “Relaxing to Charge Up Your Memory,” earlier in this chapter, for more information.

Sedating your insomnia

If you experience insomnia, you're not alone. Approximately95 percent of people report that they've had insomnia at least once in their lives. Approximately 50 percent of people report that they have trouble sleeping once a week, and about 15 percent of people report they've had trouble sleeping two nights a week.

Your sleep is very much tied to the light of the day and the dark of night. Light is taken in through your eyes, and your retina sends the information to your pineal gland, which is positioned in the middle of your brain.

Your pineal gland suppresses the production of melatonin, thereby convincing your brain that it's day-time and not time to become sedated. In contrast, when it's dark outside, your retina sends information to your pineal gland it's night outside and it should produce melatonin to induce sedation.

Watch the amount of light you're exposed to during the daytime because it affects your sleep. If you maximize your bright light exposure during the daytime, you set your body clock to match the natural day/night cycle of the world around you. If you've had trouble getting to sleep, expose yourself to bright light in the early morning. This way, you ensure your melatonin production is low throughout the day and your body temperature is the lowest when you sleep .

Exercising 3 to 6 hours before bedtime can also help you sleep better, because you push your heart rate up and still allow enough time for it to drop before you sleep. Even more importantly, you raise your body temperature and allow it to fall in time for bedtime.

If you suffer from insomnia, don't use your computer in the late evening, because it worsens your inability to sleep. When you look at the computer screen for extended periods, you're essentially looking straight at light. This light tricks your brain into adjusting to a day-time pattern because it suppresses your pineal gland's ability to secrete melatonin, which is needed to be able to go to sleep.

Chapter 22: Putting it Together – How to Memorize a Pack of Cards

Imagine being able to memorise a pack of cards as quickly as they are dealt. Sound impossible? With the right techniques, you can master this and eventually wow your friends and family. All you need is a mental map and a celebrity matrix so effective and familiar to you that you can quickly make up a memory journey right on the spot.

Learning the cards

There are 52 pieces in a deck of cards. The first step is to memorise all of them. But not in a way that you memorise each number and their corresponding symbols. You, instead, convert them into celebrities.

Remember that the mind ignores the mundane but retains the crazy and absurd. To make the association really pop out, assigning celebrities to each suit is the easiest way of recalling a deck of cards. And to make it even easier, make each suit correspond to a personality trait. Hearts can be celebrities you like, diamonds represent the super-rich, clubs for tough or crazy people, and spades for amusing or absurd people.

Once the suits are all assigned, it is now time to assign the cards to professions. Make all even number females and all odd numbers males. Make sure they are all paired up. So, for example, you can make 5 powerful men, while 6 all-powerful females, and so on.

Here's a sample of how the assigned cards would look like:

- ✓ King and Queen will each represent the male and female parts of a celebrity couple with "royal" status.
- ✓ Jack represents religious figures, as they were mostly bachelors. Just like the Jack is.
- ✓ The highest numbers, 10 and 9, can represent some of the most powerful men and women.
- ✓ 8 and 9 are the men and women who have great physiques.
- ✓ Make 6 and 5 represent some of the most controversial male and female celebs.

- ✓ Movie stars represent 4 and 3.
- ✓ 2 for all women athletes, while Ace is for men athletes.

With these assigned professions, one can then use your existing knowledge to fill up the cards with associated names. A male athlete (ace) who got rich (diamonds) could be Michael Jordan (Ace of Diamonds). Once you have the associations ready, take time to memorise and translate them into images until it becomes automatic.

Create a Memory Palace

You will need to remember 52 spots in your memory palace, and to break this down and make it easier to remember, choose 5 rooms with 10 pieces of furniture each. You can assign your bedroom, bathroom, living room, dining room, and office for this. The 10 pieces of furniture represent things that you might find in that room, and add 2 more pieces of furniture to one room because a deck of cards has 52 pieces.

Going through a deck of cards

Now, the fun begins. Once you have both your memory palace and celebrity matrix memorized fully, you can begin. Just make sure all your associations are 100% ingrained in your brain. Go through a shuffled deck of cards. For every ten cards, assign actions between the associated celebrity to the 10 pieces of furniture of the room. Then move on to the next rooms. For example, your first room is the dining room, and the first furniture is the microwave. Your first card is an Ace of Diamonds, so you can think of Michael Jordan preparing a microwave meal after a game. Assign each character at the position in which they are drawn. Continue to make the associations to the rest of the furniture before moving on to the next room. By segregating the rooms with only 10 pieces of furniture each, you won’t get so overwhelmed with the number of objects you need to memorise in one room.

Recalling and retrieving

Now that you have the stories imagined, remember what each suit and number mean and go through each room in the order of the furniture you’ve memorized. By then, you will have been able to enumerate the order of the shuffled deck you’ve just seen.

Practice makes perfect

The system works, but only with constant practice. Sharpen your mind and try to improve your speed.

Keep it fun

Whether you have plans to use this trick for competition or for amusement, remember to have fun. Retrain, practice, and perform every day until you get better, faster, and the associations become second nature.

Memory for Competitions

Memory athletes wouldn't be where they are today if not for constant and rigorous mental practices over the course of time. Just as physical sports need training, so do mental athletes need exercises. You may eventually want to remember a deck of cards as they are dealt, or memorise a sequence of numbers, or pick out a familiar face in a sea of people. If you plan to develop your memory and expand it to be able to compete, all you really need to do is work daily on the memory techniques presented in this book. Memory champions have used the method of loci and memory palace to enhance associations and linkages. As you progress, you can make these linkages even more compressed, in order to be able to memorise more. For example, let's go back to the deck of card association. Ace of Spades is represented by Michael Jordan. Next time you see this card, imagine Michael Jordan (Ace of Spades) dunking his basketball on the microwave, wearing a jersey with Queen Elizabeth's face (Queen of Hearts), eating the meat dress of Lady Gaga (Six of Spades). As you work on integrating more and more associations, you will become more adept at remembering, bringing you up to speed and improvement.

In a rapidly changing world like the one we're in now, it's necessary to adapt to the dynamic changes we face on a day-to-day basis. However, the untrained memory of a person may find it exhausting and perhaps overwhelming to remember all the significant information that happens in a single day. There are too many appointments, daily tasks, special dates, reminders, shopping lists, people to call, and other things going on in our busy minds. Because these tasks seem endless, the mind has to find a way to adapt to this very rapid shift of attention. While some can multitask, most of us suffer the inconvenience of attention deficiency and lack of focus,

which then leads to a less functional and productive human being.

Why do we forget in the first place? This is completely human in nature, and it shouldn't be treated as something embarrassing or disappointing. We forget so our minds won't get filled up with unimportant things. We forget so that it won't be hard for us to retrieve and recall old memory and data at will. Forgetting trivial information may help you retain only the most important and vital parts of information.

Memory loss can occur when a person experiences forgetfulness in names, significant dates, recent conversations, telephone numbers, and things you have recently learned. It can also include forgetting things as soon as you enter a room, misplacing things, or immediately forgetting why you entered a room in the first place. Some causes of memory loss are alcohol and drug related, such as illicit drug abuse. Other reasons are depression, which can lead to lack of focus and attention that affects memory. A head injury or a hard blow to the head can cause head trauma. Apart from physical injury, emotional trauma is also a big reason for memory loss. In worst cases, extreme stress can cause a condition called psychogenic amnesia. This makes the subject wander around and do things without them remembering what they have done. Illnesses such as HIV, tuberculosis, and herpes can affect the neurological function of our brain. Likewise, medications such as those used for sleeping, over-the-counter anti-histamines, antidepressants, and anti-anxiety meds used for schizophrenia can all affect memory. Sleep deprivation, insomnia, nutritional deficiencies, thyroid disfunction, and aging all contribute to further memory loss.

There will be times when even with consistent practice, you tend to falter and forget things. This is normal and part of being human, but there are ways you can cope in order to combat this setback and minimize future embarrassment or frustration.

When you are in the thick of using any memory palace technique, try not to multi-task. Doing so many things at the same time could lead to forgetfulness, and it is wise to perform tasks one at a time, rather than juggling everything together all at once. When trying to remember things, finish one task first. Make sure it is properly stored for effective memory retrieval later, then proceed to the next set of tasks. Conventional knowledge tells us that people who multi-task are more productive than

those who don’t. However, recent research tells us otherwise. Switching from one task to another makes it even more challenging to avoid distractions and the focus at hand. Multi-tasking is known to reduce productivity by 40% and can cause mental blocks, slowing down anyone’s productivity.

Another tip to combat forgetfulness is to repeat what someone says back to them. By doing this, your mind is taking an active part in memory making, rather than just being a passive observer. This helps strengthen muscle memory because the sheer effort in repeating things back to the person or out loud makes it simply hard to forget afterwards. Absent-mindedness is a cognitive condition wherein the person demonstrates a low level of attention, forgetfulness, and frequent distraction. In cases of absent-mindedness, the affected individual is experiencing consequences of divided attention. This is a condition where your brain is attending to more than one source of information, causing impairment to the quality of retaining information. Your mental effort, or your brain’s limited attention resources, may be impaired once you focus on more than complicated tasks. When this capacity is less than what is required for a certain task, failure tends to occur. And because this capacity is limited, it is necessary to regulate its demands.

Place an order to your system and be organized at all times as much as possible. Do not merely hear, but instead, try to listen. Scrutinize the minute details of the task at hand, and understand the concept. Blind memorization will eventually lead to deterioration of that specific memory recall. But when you understand it thoroughly, it will be very difficult to take it away from you. In this case, humans can do better when they focus their attention to one task at a time. This is called selective attention. Instead of dividing the attention among many inputs at once, it is best to concentrate on one thing at once to ensure quality storage and retention of information.

Doing spaced repetition is also another helpful way to improve memory. For example, it is easier to remember something you read yesterday than something you have read the year before. This is referred to as the forgetting curve. Research tells us that we mostly forget newly acquired information after just a few hours or at the most, days. But if you reinforce what you have just learned at regular intervals, it is easier to retain that piece of information from the long-term storage area of your brain. It all

about practicing how to remember at the right time. A simple way to reinforce this is through flashcards. If you remember something easily in a limited period of time, try to test yourself again at a much longer interval. The trick is to always reinforce the information so that you can make sure it is stored for the long haul.

Another trick that most memory palace practitioners use is the technique of chunking. This is when you club together ideas or concepts together into groups. For example, in order to remember your grocery list, you can try to list according to each shelf in the store. Or when learning a new language, group words that you will be learning by those that are related by a strong context, such as summer clothing or dinner items. Our brains have a natural tendency to look for patterns, so chunking information allows the brain to store information in easy-to-remember packets.

Simple lifestyle changes can also influence the way you can easily recall information. Taking adequate rest and sleep, consumption of brain-boosting food – these are all very important for improved memory. Exercise is also a great way to enhance memory, because it increases blood flow to the brain, leading to several cognitive benefits such as alertness, better concentration, and a more positive mood. Exercise also triggers the release of a protein called Cathepsin B. This helps in the growth of neurons and forms new connections in the hippocampus, the section of the brain playing a vital role in memory. You don’t have to undergo a rigorous exercise routine to enhance memory. A weekly 150 minute-walk is just enough to do this.

Research also tells us that socialization is known to improve memory. Better relationships are linked to happiness and improvement in a number of health parameters. It’s a great way to prevent degenerative memory conditions such as dementia and Alzheimer’s disease.

People who regularly do mentally challenging work are less likely to develop Alzheimer’s. This is because working on things that are mentally taxing keeps your neurons on their best toes and prevent them from deteriorating over time. Try to add more challenging responsibilities to your daily routine so that your cognitive abilities stay in peak shape.

Conclusion

Memory is our ability to remember everything in our daily life. Everything in your life is improved when you improve your memory. You can access information more quickly and more easily and that creates greater opportunities for connections and associations.

Perhaps Sherlock Holmes was a superhuman detective, with his vast array of mind abilities and powerful memory. But if there's anything that you can take from this book, it's the knowledge that you too can enhance and improve your memory. Whether it's for casual or competition purposes, memory enhancement techniques are very effective at helping you unlock a whole new potential. Apart from impressing your friends and family, the improved focus that comes along with it can continue to stimulate your brain, making it age less fast. Remember that memories are all that we take when all is said and done, and if we let it deteriorate, it can be very devastating. Memory enhancement techniques also serve to keep our brains young and happy, so do your brain a favor and keep it up to its toes in stimulation. You don't need to be Sherlock Holmes, but you can be your own memory champion.

-- Dane Krauss

Dear Reader,

Thanks for exploring this book with me. Now that you know how to develop a photographic memory…

…why not take one step further and learn about the intriguing world of hypnosis?

You'll love the other book on the improvement of the mind, because it complements this one.

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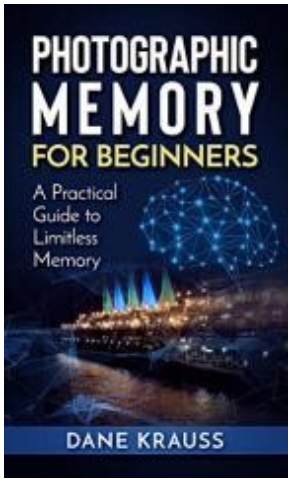
Thanks,

Dane

P.S. Reviews are like giving a warm hug to your favorite author. We love hugs.



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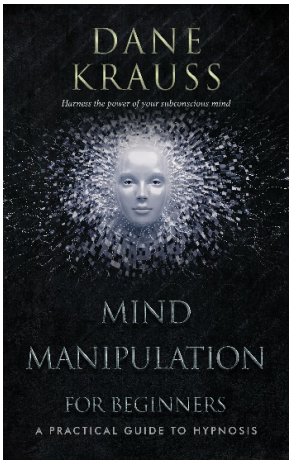

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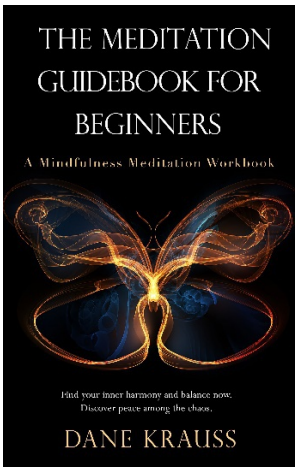

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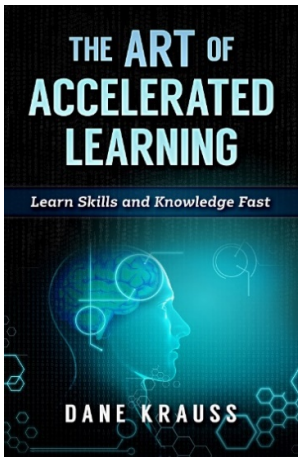

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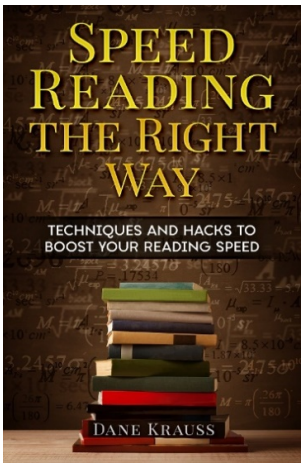

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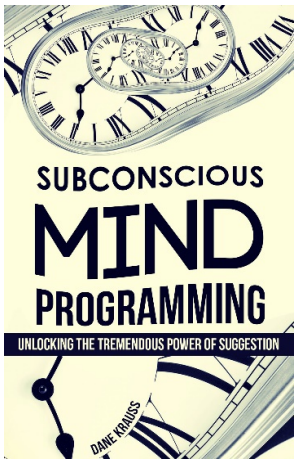

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